Started on	Sunday, 24 July 2022, 6:00 PM
State	Finished
Completed on	Sunday, 24 July 2022, 6:00 PM
Time taken	9 secs
Marks	0.00/229.00
Grade	**0.00** out of 100.00

Question **1**

Not answered

Receptors and side effects

Identify the receptors responsible for each of the following side effects induced by psychotropic medications

Weight gain caused by antipsychotics	Choose...
Hyperprolactinaemia induced by antipsychotics	Choose...
Drug-induced parkinsonism caused by antipsychotics	Choose...
Gastrointestinal discomfort caused by SSRIs	Choose...
Postural hypotension induced by antipsychotics	Choose...

Choose...
5HT2C stimulation
D-2 blockade
Alpha-1 antagonism
5HT2A blockade
Alpha-1 stimulation
5HT2A stimulation
5HT3 stimulation
5HT-1D stimulation
GABA-A stimulation
5HT2C Antagonism

Explanation:

Increased weight gain associated with some antipsychotics may be due to actions at the H1 histamine receptor and the 5HT2C serotonin receptor. When these receptors are blocked, particularly at the same time, patients can experience weight gain.

When D2 antagonists are administered, D2 receptors in the tuberoinfundibular dopamine pathway are blocked. This causes plasma prolactin concentrations to rise, a condition called hyperprolactinaemia.

Motor side effects are caused by D2 antagonists/partial agonists blocking D2 receptors in the nigrostriatal motor pathway. When D2 receptors are blocked acutely in the nigrostriatal pathway, this can cause drug-induced parkinsonism (DIP). Often, any abnormal motor symptoms caused by D2 receptor blockers are lumped together and called collectively extrapyramidal symptoms (EPS), but EPS is an old-fashioned and relatively imprecise term for describing the motor side effects of D2 antagonists/partial agonists.

5HT3 receptors are localised in the chemoreceptor trigger zone of the brainstem, where they mediate nausea and vomiting (especially in response to cancer chemotherapy), and in the gastrointestinal tract itself where they mediate nausea, vomiting, and diarrhea/bowel motility when stimulated by serotonin (including when stimulated by serotonin that is a side effect of peripherally increased serotonin by SSRIs/SNRIs). Blocking these 5HT3 receptors can therefore protect against chemotherapy-induced nausea and vomiting as well as against serotonin-induced gastrointestinal side effects that can accompany agents that increase serotonin.

Postural hypotension is commonly associated with antipsychotics drugs that are antagonists at postsynaptic alpha-1-adrenergic receptors.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 164-166, 198, 309-311.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 161.

The correct answer is: Weight gain caused by antipsychotics → 5HT2C Antagonism, Hyperprolactinaemia induced by antipsychotics → D-2 blockade, Drug-induced parkinsonism caused by antipsychotics → D-2 blockade, Gastrointestinal discomfort caused by SSRIs → 5HT3 stimulation, Postural hypotension induced by antipsychotics → Alpha-1 antagonism

Question **2**

Not answered

Marked out of 4.00

Drugs and adverse effects

For each of the given adverse effect that leads to restricted use and requirements for appropriate monitoring, choose the most likely offending agent.

Seizures	Choose...
Ovarian cyst	Choose...
Steven Johnson syndrome	Choose...
QTc prolongation	Choose...

Choose...
Sodium Valproate
Olanzapine
Ziprasidone
Carbamazepine
Haloperidol
Acamprosate
Methylphenidate
Risperidone
Bupropion
Lithium

Explanation:

Bupropion has a dose-related risk of seizures (particularly with instant-release formulations). Risk is less with slow-release formulations at doses under 300mg/day.

Valproate can cause hyperandrogenism in women and has been linked with the development of polycystic ovaries. Teratogenicity and polycystic ovary syndrome with associated infertility are particular concerns. Valproate should not be used in women of child-bearing age.

Carbamazepine can cause Steven Johnson syndrome and toxic epidermal necrolysis (TEN). Individuals with the HLA-B*1502 allele are at increased risk of Steven Johnson syndrome and TEN.

Ziprasidone prolongs the QTc interval more than some other antipsychotics. Ziprasidone is contraindicated in patients with a known history of QTc prolongation, recent acute myocardial infarction, and uncompensated heart failure.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 259, 744, 789.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 134, 86

The correct answer is: Seizures → Bupropion, Ovarian cyst → Sodium Valproate, Steven Johnson syndrome → Carbamazepine, QTc prolongation → Ziprasidone

Question **3**

Not answered

Marked out of 3.00

Flag question

Psychiatric side effects of non-psychiatric medications

For each of the side effects profile given below identify the possible offending drug

A 74-year-old gentleman being treated for resting tremors develops acute confusional state with paranoia and visual hallucinations	Choose...
A 67-year-old lady taking treatment for swollen legs complains of visual halos, which are green in colour and appears to be confused	Choose...

Choose...
Amiodarone
Ramipril
Bendroflumethiazide
Digoxin
Alpha Methyldopa
Levodopa
Prednisolone
Lithium
Fluoxetine
Isoniazid

A 56-year-old lady on treatment for polymyalgia rheumatica becomes hypomanic and confused — Choose...

Explanation:

The patient in scenario 1 is likely to be taking treatment for Parkinson's disease using levodopa. Levodopa has been associated with visual hallucinations, depression, hypomania, sleep disturbances, abnormal dreams, cognitive impairment, agitation, psychosis, and delirium.

The patient in scenario 2 is likely to be suffering from cardiac failure, and is taking digoxin. One of the side effects of cardiac glycosides such as digoxin is a change in vision characterized by seeing a halo or light around objects.

The patient in scenario 3 is likely to be taking steroids for her autoimmune condition. There is a clear causal association of glucocorticoids (e.g. betamethasone, prednisolone, prednisone) with psychiatric side effects, including mood disorders, suicidal ideation, euphoria, agitation, sleep disturbances, psychosis, delirium, and cognitive impairment.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 918-919.

The correct answer is: A 74-year-old gentleman being treated for resting tremors develops acute confusional state with paranoia and visual hallucinations → Levodopa, A 67-year-old lady taking treatment for swollen legs complains of visual halos, which are green in colour and appears to be confused → Digoxin, A 56-year-old lady on treatment for polymyalgia rheumatica becomes hypomanic and confused → Prednisolone

Question **4**

Not answered

Marked out of 3.00

Choose...
Fluoxetine
Risperidone
Paroxetine
Duloxetine
Venlafaxine
Citalopram
Childhood trauma
Mirtazapine
Trazodone

Sexual side effects
For each of the following side effects, choose the most likely associated cause from the given list

A 34-year-old patient presents to an emergency unit with painfully prolonged erection. He is being treated for low mood and insomnia. — Choose...

A 41-year-old man complains of decreased sexual interest, as he is not able to ejaculate normally. He has been maintained on a treatment for chronic auditory hallucinations. — Choose...

A 32-year-old woman complains of new onset pain during intercourse. She recently started treatment for depression. — Choose...

Explanation:

In rare cases, trazodone causes priapism. Trazodone should be discontinued if prolonged penile erection occurs because of the risk of permanent erectile dysfunction. Patients should seek medical attention immediately if painful erections occur lasting more than an hour. Generally, priapism reverses spontaneously, while penile blood flow and other signs are being monitored, but in urgent cases, local phenylephrine injections or even surgery may be indicated.

Specific peripheral alpha-1 adrenergic blockade with risperidone leads to a moderately high reported incidence of ejaculatory problems such as retrograde ejaculation. Prevalence of sexual dysfunction with risperidone is reported to be 60-70%.

Paroxetine has been associated with a higher incidence of sexual dysfunction than other SSRIs and antidepressants. Data on the effect of antidepressants on sexual pain are mixed, with some studies suggesting lubrication problems leading to pain with vaginal penetration, some reporting no effect, and

still others reporting improvements in vulvodynia with antidepressant use.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 795.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 174.

Lorenz T, Rullo J, Faubion S. Antidepressant-induced female sexual dysfunction. InMayo Clinic Proceedings 2016 Sep 1 (Vol. 91, No. 9, pp. 1280-1286). Elsevier.

The correct answer is: A 34-year-old patient presents to an emergency unit with painfully prolonged erection. He is being treated for low mood and insomnia. → Trazodone, A 41-year-old man complains of decreased sexual interest, as he is not able to ejaculate normally. He has been maintained on a treatment for chronic auditory hallucinations. → Risperidone, A 32-year-old woman complains of new onset pain during intercourse. She recently started treatment for depression. → Paroxetine

Question **5**

Not answered

Marked out of 3.00

Teratogenicity of psychotropics
Choose the most likely offending agent for each of the following teratogenic effects from the given list.

Anencephaly — Choose...

Defect of the tricuspid valve — Choose...

Spina bifida — Choose...

Choose...
Lamotrigine
Valproate
Reboxetine
Zopiclone
Mirtazapine
Carbamazepine
Venlafaxine
Paroxetine
Lithium
Diazepam

Explanation:

Valproate is associated with strong evidence for increased risk of congenital and major congenital malformations (as many as 11% of exposed children); cleft lip and/or palate, neural tube defects (risk of spina bifida 1-2%), congenital heart anomalies, radial ray defects, ophthalmological and genitourinary anomalies; developmental delay (delayed walking and talking, memory problems, difficulty with speech and language), low IQ, and autism. This risk increases with increasing doses of valproic acid, especially when the dose is more than 1gm/day.

Much attention has been focused on the rare but severe cardiac anomaly of the Ebstein type. There may be an increased risk of **Ebstein anomaly** (prevalence in the **general population** of 1:20 000) if lithium taken during 1st trimester (maximum risk 2-6 weeks after conception), though the size of risk uncertain. More recent data suggest the magnitude of effect is **much smaller** than previously estimated. Recent data have shown an **association with maternal mental health problems** generally rather than specifically with lithium.

Ref: McAllister-Williams RH et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. Journal of Psychopharmacology. 2017 May;31(5):519-52.

National Institute for Health and Clinical Excellence. Antenatal and postnatal mental health: clinical management and service guide. Updated edition 2020.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 690-691.

The correct answer is: Anencephaly → Valproate, Defect of the tricuspid valve → Lithium, Spina bifida → Valproate

Question **6**

Not answered

Choose...
Mianserin
Fluvoxamine
Fluoxetine
Mirtazapine
Paroxetine
Venlafaxine
Agomelatine
Citalopram
Reboxetine
Escitalopram

Drug Interactions
Identify the antidepressant drug that is most likely to interact and should be avoided in the following situations

A 33-year-old woman treated for mixed anxiety & depressive disorder. She is on diazepam, started by her GP. Choose...

A 21-year-old asthmatic on theophylline is suffering from depression. Choose...

A 67-year-old gentleman with a previous history of stroke and is now depressed. He is on regular warfarin. Choose...

Explanation:

Fluvoxamine is a potent inhibitor of CYP1A2, and to a lesser extent of CYP2C and CYP3A4, and has a high potential for interactions. Via CYP3A4 inhibition, fluvoxamine may reduce clearance of benzodiazepines such as diazepam. Because fluvoxamine is a potent inhibitor of CYP1A2, it can cause increased theophylline serum levels. Fluvoxamine may displace highly protein-bound drugs, such as warfarin, and can cause an increased risk of bleeding.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 308, 341.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 328.

The correct answer is: A 33-year-old woman treated for mixed anxiety & depressive disorder. She is on diazepam, started by her GP. → Fluvoxamine, A 21-year-old asthmatic on theophylline is suffering from depression. → Fluvoxamine, A 67-year-old gentleman with a previous history of stroke and is now depressed. He is on regular warfarin. → Fluvoxamine

Question **7**

Not answered

Marked out of 4.00

Choose...
Topiramate
Lamotrigine
Quetiapine
Olanzapine
Fluoxetine
Lorazepam
Risperidone
Valproate
Lithium

Side effects of treating bipolar disorder
The following patients with bipolar disorder have all presented with side effects. Please choose the most likely offending agent from the list shown here.

A 38-year-old man comes to see you with severe central abdominal pain. It is associated with vomiting. An amylase level is highly elevated. Choose...

A 52-year-old woman complains of thirst. She is also passing a lot of urine. On examination, she is well hydrated but has a fine tremor. Choose...

A 25-year-old man with bipolar disorder starts on a second drug. The dose is increased quickly. He develops a rash on his skin and blistering on his lips and gums and a rash around his eyes. On examination, you find target-like lesions on his trunk. Choose...

A 40-year-old man comes to see you, as he is worried about his libido. He is also concerned as he thinks he is developing breasts. On examination, you find gynaecomastia. Choose...

Explanation:

Rarely, valproate can cause pancreatitis, which is sometimes fatal.

Polyuria due to impaired urinary concentrating ability occurs in up to 20% of patients chronically treated with lithium. Adverse effects of lithium also include gastrointestinal upset and fine tremor.

10-14% of patients receiving lamotrigine will develop rash. Most are benign. A minority may be serious/life-threatening skin reactions requiring hospitalisation (e.g. Steven-Johnson syndrome, toxic epidermal necrolysis, angio-oedema, and a rash associated with a number of systemic manifestations). Rash is most likely to occur within the first 2-8 weeks of treatment and is more likely when combined with valproate, exceeding the recommended initial dose, or rapid dose escalation.

Hyperprolactinaemia caused by antipsychotics can in turn cause secondary amenorrhea in women and breast enlargement and galactorrhea in both men and women. Amisulpride, sulpiride, risperidone, and paliperidone are particularly associated with hyperprolactinaemia.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 830

Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. pp. 270.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 39, 168, 250.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 358.

The correct answer is: A 38-year-old man comes to see you with severe central abdominal pain. It is associated with vomiting. An amylase level is highly elevated. → Valproate, A 52-year-old woman complains of thirst. She is also passing a lot of urine. On examination, she is well hydrated but has a fine tremor. → Lithium, A 25-year-old man with bipolar disorder starts on a second drug. The dose is increased quickly. He develops a rash on his skin and blistering on his lips and gums and a rash around his eyes. On examination, you find target-like lesions on his trunk. → Lamotrigine, A 40-year-old man comes to see you, as he is worried about his libido. He is also concerned as he thinks he is developing breasts. On examination, you find gynaecomastia. → Risperidone

Question **8**

Not answered

Marked out of 4.00

Drug Toxicity (1)
Choose the most important treatment option for the drug-induced toxic symptoms given below

An 83-year-old woman presents with signs of lithium toxicity including disorientation, seizures, and hypotension. — Choose...

A 23-year-old gentleman who took an overdose of methadone prescription and has pinpoint pupils. — Choose...

A 56-year-old gentleman who is on phenelzine therapy consumes large quantities of cheese and wine and has a very high blood pressure. — Choose...

A 56-year-old gentleman with depression took a dangerous overdose of barbiturates needs additional treatment to enhance drug clearance. — Choose...

Choose...
Diazepam
Peritoneal dialysis
Phentolamine
Naloxone
Thiopentone
Alkalinisation of urine
Rehydration
Propranalol
Haemodialysis
Flumazenil
Acidification of urine

Explanation:

Toxic effects of lithium reliably occur at levels >1.5 mmol/L and usually consist of gastrointestinal effects (increasing anorexia, nausea, and diarrhoea) and CNS effects (muscle weakness, drowsiness, confusion, ataxia, course tremor and muscle twitching). Above 2 mmol/L, increased disorientation and seizures usually occur, which can progress to coma, and ultimately death. In the presence of more severe symptoms, osmotic or forced alkaline diuresis should be used (NB never thiazide or loop diuretics). Above 3 mmol/L, peritoneal dialysis or haemodialysis is often used.

Naloxone is an opioid receptor antagonist that can reverse opioid overdose.

The most worrisome side effect of MAOIs is the tyramine-induced hypertensive crisis ('cheese reaction'). Sources of dietary tyramine include cheese, meat extracts and yeast extracts, alcohol, non-fresh fish, non-fresh poultry, offal, avocado, banana skins, broad bean pods, caviar, and herring (pickled or smoked). An MAOI-induced hypertensive crisis should be treated with alpha-adrenergic

antagonists (for example phentolamine or chlorpromazine).

In the setting of a barbiturate overdose, alkalinisation of the urine increases excretion.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 250-251, 474.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 645-646.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 282.

The correct answer is: An 83-year-old woman presents with signs of lithium toxicity including disorientation, seizures, and hypotension. → Haemodialysis, A 23-year-old gentleman who took an overdose of methadone prescription and has pinpoint pupils. → Naloxone, A 56-year-old gentleman who is on phenelzine therapy consumes large quantities of cheese and wine and has a very high blood pressure. → Phentolamine, A 56-year-old gentleman with depression took a dangerous overdose of barbiturates needs additional treatment to enhance drug clearance. → Alkalinisation of urine

Question **9**

Not answered

Marked out of 4.00

Drug Toxicity (2)

Choose an appropriate treatment option for the toxic symptoms produced by psychotropic drugs

A 66-year-old woman who took an overdose of aspirin and presents with altered levels of consciousness requires additional treatment to enhance drug clearance. Choose...

A 21-year-old drug addict consumed high-dose amphetamine in a nightclub accidentally and went to a state of coma. He has hyperthermia on presentation. Choose...

An 18-year-old girl presented to A&E after consuming an overdose of diazepam tablets and presents with drowsiness and confusion. Choose...

A 56-year-old woman took an overdose of amitriptyline due to chronic pain presented to A&E with hyperreflexia, myoclonic twitches and altered levels of consciousness. Choose...

Choose...
Evaporative cooling
Flumazenil
Diazepam
Naloxone
Propranalol
Haemodialysis
Phentolamine
Acidification of urine
Peritoneal dialysis
Alkalinisation of urine
Thiopentone

Explanation:

Aspirin & amphetamines: the ionized forms are more or less trapped in the glomerular filtrate and do not get reabsorbed. Hence, renal clearance is more for ionized drugs. A weak base can be ionised by acidifying urine; a weak acid by alkalinising urine. Hence for salicylate (aspirin) overdose, and barbiturate overdose, alkalinisation helps.

Urine alkalinisation to a pH of 7.5-8.0 increases urinary excretion of salicylates more than 10-fold and should be considered for significant salicylate toxicity in patients with intact renal function, alone or in combination with haemodialysis. Intravenous (not oral) administration of sodium bicarbonate as a crystalloid preparation should be used.

Acidification may help in the elimination of amphetamines and phencyclidine, but often complications of such a procedure (**myoglobin precipitation in the renal tubules)** override any benefits. Aggressive cooling of hyperthermic patients with evaporative cooling, ice packs to the groin and axilla may be required to prevent multiorgan failure. Some centers employ "ice-bath" to this effect.

Diazepam: Benzodiazepines are not considered fatal on their own even in cases of a considerable overdose. Nevertheless, diazepam overdose is a medical emergency. If respiratory depression or cardiovascular complications are present, then an antagonist for benzodiazepines called flumazenil can be used. Unlike diazepam that stays in the body for days, flumazenil is a short-acting drug (41-79 minutes), and thus multiple doses of flumazenil may be required to reverse the effect of overdose fully.

Amitriptyline in higher doses can cause hypotension, hyperreflexia, and QT prolongation and

precipitate seizures. Tricyclic antidepressants impose their therapeutic effects by inhibiting presynaptic reuptake of norepinephrine and serotonin in the central nervous system (CNS). This effect in the CNS can cause seizures. Hyperreflexia and myoclonic twitches indicate an increased risk for seizures. Seizures usually respond to benzodiazepines, such as diazepam. In the case of refractory seizures, prompt administration of anticonvulsants, such as phenobarbital or propofol, or even general anaesthesia, should be considered.

Ref: Guidance document: management priorities in salicylate toxicity. Journal of Medical Toxicology. 2015 Mar;11:149-52.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 294.

The correct answer is: A 66-year-old woman who took an overdose of aspirin and presents with altered levels of consciousness requires additional treatment to enhance drug clearance. → Alkalinisation of urine, A 21-year-old drug addict consumed high-dose amphetamine in a nightclub accidentally and went to a state of coma. He has hyperthermia on presentation. → Acidification of urine, An 18-year-old girl presented to A&E after consuming an overdose of diazepam tablets and presents with drowsiness and confusion. → Flumazenil, A 56-year-old woman took an overdose of amitriptyline due to chronic pain presented to A&E with hyperreflexia, myoclonic twitches and altered levels of consciousness. → Diazepam

Question **10**

Not answered

Marked out of 5.00

Clozapine-related adverse effects

For each of the following side effects caused by clozapine, identify the best plan of action

Choose...
Increase the dose
Perform EEG
Hyoscine 300 micrograms at night time
Recommend high fibre diet
Give smaller dose in the morning
Check full blood count
Consider prophylactic valproate treatment
Avoid fluids before bed time
Titrate the dose upwards quickly
Dietary counselling
Advise gradual postural change

A 34-year-old woman started on clozapine suffers from high temperature	Choose...
A 65-year old gentleman with one episode of seizures ten years ago now requires high dose of clozapine to treat psychosis	Choose...
A 33-year-old woman has gained weight on clozapine after taking it for three months	Choose...
A 45-year-old woman complains of excessive salivation especially at night time	Choose...
A 56-year-old woman recently started on clozapine complaints of dizzy spells on waking up from bed	Choose...

Explanation:

Clozapine induces an inflammatory response (increased CRP, interleukin-6, and eosinophils). In the setting of fever, check a full blood count for neutropenia. Also beware of myocarditis, neuroleptic malignant syndrome, pneumonia, and other rarer types of inflammatory organ damage.

Consider prophylactic topiramate, lamotrigine, gabapentin, or valproate if patients are on high dose clozapine (>500mg/day) or with high plasma level (>500ug/L).

In the setting of weight gain, dietary counselling is essential. Advice may be more effective if given before weight gain occurs.

Clozapine-induced hypersalivation is often very troublesome at night. It can be addressed by giving hyoscine 300ug sucked and swallowed.

If orthostatic hypotension occurs, advise the patient to take time when standing up.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th

Edition. 2021. pp. 212-213.

The correct answer is: A 34-year-old woman started on clozapine suffers from high temperature → Check full blood count, A 65-year old gentleman with one episode of seizures ten years ago now requires high dose of clozapine to treat psychosis → Consider prophylactic valproate treatment, A 33-year-old woman has gained weight on clozapine after taking it for three months → Dietary counselling, A 45-year-old woman complains of excessive salivation especially at night time → Hyoscine 300 micrograms at night time, A 56-year-old woman recently started on clozapine complaints of dizzy spells on waking up from bed → Advise gradual postural change

Question **11**

Not answered

Marked out of 6.00

Plasma levels

For each of the following drugs, identify the most appropriate range of target plasma levels.

Sodium valproate	Choose...
Lithium	Choose...
Carbamazepine	Choose...
Clozapine	Choose...
Phenytoin	Choose...
Amitriptyline	Choose...

Choose...
200-320 micrograms/L
0.6-1 mmol/L
50-100 mg/L
20-60 micrograms/L
More than 7 mg/L
150-210 micrograms/L
50-150 micrograms/L
350-500 micrograms/L
100-200 micrograms/L
10-20 mg/L

Explanation:

Clozapine: 350-500 micrograms/L.

Carbamazepine: More than 7 mg/L.

Lithium: 0.6-1 mmol/L.

Sodium valproate: 50-100 mg/L. Uncertainty exists over the value of levels in epilepsy and bipolar disorder. Some evidence that levels up to 125 mg/L are tolerated and more effective than lower levels in mania.

Phenytoin: 10-20 mg/L. Phenytoin follows zero-order kinetics. Free rather than plasma-bound levels of phenytoin may be useful in some circumstances.

Amitriptyline: 100-200 micrograms/L.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 733-734.

The correct answer is: Sodium valproate → 50-100 mg/L, Lithium → 0.6-1 mmol/L, Carbamazepine → More than 7 mg/L, Clozapine → 350-500 micrograms/L, Phenytoin → 10-20 mg/L, Amitriptyline → 100-200 micrograms/L

Question **12**

Not answered

Marked out of 5.00

Flag

Mechanism of action

Select the primary mechanism of action for each of the following drugs from the list shown here.

Ramelteon	Choose...

question

Choose...
GABA-A agonist
Presynaptic alpha-2 agonist
MT1/MT2 receptor agonist
MT3/MT4 receptor agonist
Partial agonist at the α4β2 unit of nicotinic acetylcholine receptor
D-2 receptor partial agonist
5 -HT1-a partial agonist
MAO-B Inhibitor
NMDA glutamate receptor antagonist
Selective presynaptic norepinephrine reuptake inhibition

Riluzole	Choose...
Atomoxetine	Choose...
Clonidine	Choose...
Varenicline	Choose...

Explanation:

Ramelteon: This is a MT1/MT2 receptor agonist promoted as a hypnotic. It decreases sleep latency and increases sleep time across all ages; the dose-response curve is flat with no significant difference in efficacy between the 16-mg or 64-mg doses. Hence, it may have lower abuse potential than other hypnotics but its endocrinological effects need to be fully evaluated yet.

Riluzole was developed to slow the progression of amyotrophic lateral sclerosis (ALS, or Lou Gehrig disease). Theoretically, riluzole binds to voltage-sensitive sodium channels and prevents glutamate release, in an action similar to that postulated for lamotrigine. The idea is that diminishing glutamate release would prevent the postulated excitotoxicity that may be causing the death of motor neurons in ALS. Riluzole also has other effects such as blockade of high-voltage calcium channels and NMDA-glutamate receptor antagonism. It is unclear which of these properties contribute most to its behavioural effects in therapeutic use.

Atomoxetine is a selective norepinephrine reuptake inhibitor (NRI). Blocking norepinephrine transporters (NETs) in the prefrontal cortex increases both dopamine and norepinephrine (NE) in the prefrontal cortex and is why NET inhibitors are thought to work in ADHD. However, since there are few NE neurons and NETs in nucleus accumbens, inhibiting NET does not lead to an increase in either NE or DA there and is why NET inhibitors are thought not to have reinforcing, abuse, or addiction potential.

Lofexidine & clonidine: In opioid detoxification, if a short duration of treatment is desirable, α-2 adrenergic agonists such as lofexidine and clonidine are preferred to methadone. Clonidine is a centrally acting alpha 2 agonist antihypertensive agent and a nonstimulant treatment for ADHD.

Varenicline is a partial agonist at the α4β2 unit of nicotinic acetylcholine receptor. It is shown to assist smoking cessation by relieving nicotine withdrawal symptoms and reducing the rewarding properties of nicotine.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 179 677, 837.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 352-353, 480.

The correct answer is: Ramelteon → MT1/MT2 receptor agonist, Riluzole → NMDA glutamate receptor antagonist, Atomoxetine → Selective presynaptic norepinephrine reuptake inhibition, Clonidine → Presynaptic alpha-2 agonist, Varenicline → Partial agonist at the α4β2 unit of nicotinic acetylcholine receptor

Question **13**

Not answered

Marked out of 3.00

Flag

Psychotropic induced ECG changes

Psychotropic use is often associated with ECG changes, necessitating a good working knowledge of ECG findings for a psychiatric trainee. For each of the following clinical conditions, identify the possible ECG changes.

A 65-year-old smoker with schizophrenia and

Choose...
P-R interval of 100 ms
P-R interval of 500 ms
Diffuse ST elevation with saddle formation
P waves absent before every QRS
R-R interval of 1500ms
Prolonged QRS interval
Corrected QT of 510 ms
Corrected QT of 440 ms
P-R interval of 280 ms
R-R interval of 500 ms

IDDM suffers from chest pain and shortness of breath. The heart rate is 40 per minute. In addition to a marked ST elevation in inferior leads, what other feature can be expected? Choose...

A 27-year-old woman suffers from depression and is on Amitriptyline. She develops infectious diarrhoea and collapses. ECG shows polymorphic ventricular tachycardia. What feature could be seen in her ECG recorded prior to developing tachycardia? Choose...

A 40-year-old woman is receiving Transcranial Magnetic Stimulation for resistant depression. She is also taking imipramine. Choose...

Explanation:

Question 1 describes a case of acute myocardial infarction presenting with bradycardia. ST elevation is a tell-tale sign of ischemia and inferior wall is commonly involved; RR interval is prolonged reflecting the bradycardia.

Accordig to OBrien (2003), "Tricyclic antidepressants delay cardiac conduction mainly by prolonging the second part of the QRS interval without significantly affecting the first. The effect is dose-dependent, occurring at both therapeutic and toxic doses, particularly with amitriptyline, imipramine and nortriptyline, but not doxepin". (Ref: http://apt.rcpsych.org/content/9/6/414)

TCAs (especially amitriptyline) have a moderate effect on QTc prolongation. There is some controversy over the exact association between QTc and risk of arrhythmias. Rather strong evidence links QTc values over 500 ms to an apparently increased risk of polymorphic ventricular arrhythmia (torsades de pointes).

The correct answer is: A 65-year-old smoker with schizophrenia and IDDM suffers from chest pain and shortness of breath. The heart rate is 40 per minute. In addition to a marked ST elevation in inferior leads, what other feature can be expected? → R-R interval of 1500ms, A 27-year-old woman suffers from depression and is on Amitriptyline. She develops infectious diarrhoea and collapses. ECG shows polymorphic ventricular tachycardia. What feature could be seen in her ECG recorded prior to developing tachycardia? → Corrected QT of 510 ms, A 40-year-old woman is receiving Transcranial Magnetic Stimulation for resistant depression. She is also taking imipramine. → Prolonged QRS interval

Question **14**

Not answered

Marked out of 3.00

Flag question

Adverse effects (1)

Identify the type of side effects caused by antipsychotics, using descriptions given below.

Choose...
Akathisia
Tardive dyskinesia
Serotonin syndrome
Parkinsonism
Hyponatraemia
Neuroleptic malignant syndrome
Cerebellar ataxia
Acute dystonia
Cholinergic rebound syndrome
SIADH

An 18-year-old gentleman commenced on haloperidol presents dramatically with eyes rolled upwards, and a twisted neck. Choose...

25-year-old woman on the ward commenced on risperidone is constantly pacing up and down, rocking from foot to foot and always crossing/uncrossing legs when seated. Choose...

A 33-year-old gentleman was started on trifluoperazine five days ago. He is now complaining of tremors, stiffness and inability to initiate body movement. Choose...

Explanation:

Dystonia is a syndrome of sustained, often painful muscular spasms, producing repetitive, twisting movements, or abnormal postures that develop following exposure to antipsychotic medication. Symptoms/signs include torticollis, trismus, blepharospasm, oculogyric crisis, and opisthotonus.

Akathisia is an unpleasant, distressing side effect of antipsychotic treatment. It characteristically manifests with a subjective feeling of inner restlessness and an objective component – movements such as pacing constantly; inability to stand, sit, or lie still; rocking; and crossing/uncrossing legs.

Drug-induced parkinsonism includes tremor, rigidity, and bradykinesia.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 1010, 1012, 1016.

The correct answer is: An 18-year-old gentleman commenced on haloperidol presents dramatically with eyes rolled upwards, and a twisted neck. → Acute dystonia, 25-year-old woman on the ward commenced on risperidone is constantly pacing up and down, rocking from foot to foot and always crossing/uncrossing legs when seated. → Akathisia, A 33-year-old gentleman was started on trifluoperazine five days ago. He is now complaining of tremors, stiffness and inability to initiate body movement. → Parkinsonism

Question **15**

Not answered

Marked out of 3.00

Adverse effects (2)

Identify the type of side effects, using descriptions given below.

A 36-year-old woman treated on fluoxetine and phenelzine for her depression has now developed diarrhoea, excessive sweating and myoclonus. Choose...

A 76-year-old gentleman admitted to hospital is on paroxetine for more than five years. He has not taken his antidepressant medication for last five days. He has developed flu-like symptoms, paresthesias and complains of electric shock-like sensations. Choose...

A 73-year-old woman started on fluoxetine recently was noticed by her CPN to be confused and is complaining of dizziness, nausea and lethargy. Choose...

Choose...
Acute dystonia
Parkinsonism
Serotonin syndrome
Antidepressant discontinuation syndrome
Cerebellar ataxia
Cholinergic rebound syndrome
Neuroleptic malignant syndrome
Tardive dyskinesia
Hyponatraemia

Explanation:

Serotonin syndrome, a potentially life-threatening acute reaction to serotonergic drugs, occurs more frequently with a combination of serotonergic drugs or in overdose. Most severe cases of serotonin syndrome involve an MAOI plus an SSRI. The onset of the syndrome is usually rapid.

SSRI discontinuation symptoms may include paresthesia, flu-like symptoms, ‘shock-like’ sensations, dizziness, insomnia, excessive dreaming, irritability, and tearfulness. The onset of symptoms is usually rapid (days, or within 3-5 half-lives of the drug). The mechanisms underly the discontinuation symptoms of antidepressants are not fully understood. After abrupt discontinuation, the sudden deprivation of serotonin coupled with temporary adaptive changes probably triggers the symptoms.

Most antidepressants have been associated with hyponatraemia. It has been suggested that serotonergic drugs are relatively more likely than noradrenergic drugs to cause hypnonatraemia, although this is

disputed. Risk factors include: older age (most important), female sex, major surgery, history of hyponatraemia or a low baseline Na concentration, co-therapy with other drugs known to be associated hyponatraemia (e.g. diuretics, NSAIDs, antipsychotics, carbamazepine, cancer chemotherapy, calcium antagonists, ACE inhibitors, and laxatives), reduced renal function, medical co-morbidity (e.g. hypothyroidism, diabetes, COPD, hypertension, head injury, congestive cardiac failure, cerebrovascular accident, various cancers), and low body weight.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 683.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 343, 393-394.

Rizkalla M et al. Antidepressant Discontinuation Syndrome: A Common but Underappreciated Clinical Problem. Journal of Osteopathic Medicine. 2020 Mar 1;120(3):174-8.

The correct answer is: A 36-year-old woman treated on fluoxetine and phenelzine for her depression has now developed diarrhoea, excessive sweating and myoclonus. → Serotonin syndrome, A 76-year-old gentleman admitted to hospital is on paroxetine for more than five years. He has not taken his antidepressant medication for last five days. He has developed flu-like symptoms, paresthesias and complains of electric shock-like sensations. → Antidepressant discontinuation syndrome, A 73-year-old woman started on fluoxetine recently was noticed by her CPN to be confused and is complaining of dizziness, nausea and lethargy. → Hyponatraemia

Question **16**

Not answered

Marked out of 5.00

Choose an appropriate drug that could be used in the management of the following conditions.

Self-injurious behaviour in people with autism spectrum disorders and intellectual disabilities not responding to antipsychotics — Choose...

Maintenance treatment for bipolar mania — Choose...

Treatment of bipolar depression — Choose...

Attention deficit hyperactivity disorder in people over 18 — Choose...

Akathisia — Choose...

Choose...
Modafinil
Cyproheptadine
Sodium valproate
Methylphenidate
Fluoxetine
Lamotrigine
Naltrexone
Esketamine
Melatonin

Explanation:

Aggression towards others and the self are common problems in autism spectrum disorder (ASD). Risperidone and aripiprazole have been reliably shown to help irritability and associated disruptive behaviours (aggression, self-injurious behaviour, hyperactivity). Naltrexone has been used for severe self-injurious behaviour, but the evidence is not strong and the results are inconsistent. Its use may still be an option in severe and intractable cases.

Both valproate and olanzapine are recommended in current practice guidelines for the treatment and prophylaxis of bipolar mania.

For the treatment of bipolar depression, NICE recommends the initial use of fluoxetine combined with olanzapine, or quetiapine on its own (assuming that an antipsychotic is not already prescribed). Lamotrigine is considered second-line. BAP guidelines have lamotrigine as a first-line option, albeit with the caveat that a mood stabiliser or antipsychotic will be needed to protect against mania.

Methylphenidate or lisdexamfetamine are recommended for use in adults with ADHD as first-line treatments.

There is limited evidence on the benefit-risk balance for any pharmacological treatment for akathisia.

Options include switching to an antipsychotic with a lower liability for the condition, or adding a beta-adrenergic blocker, a 5-HT2A antagonist (such as mirtazapine or mianserin), an anticholinergic agent, or cyproheptadine (an antihistamine).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 114, 288, 298, 571, 581, 800.

National Institute for Health and Clinical Excellence. The NICE guideline on the assessment and management of bipolar disorder in adults, children and young people in primary and secondary care. Updated edition 2020.

The correct answer is: Self-injurious behaviour in people with autism spectrum disorders and intellectual disabilities not responding to antipsychotics → Naltrexone, Maintenance treatment for bipolar mania
→ Sodium valproate, Treatment of bipolar depression → Lamotrigine, Attention deficit hyperactivity disorder in people over 18 → Methylphenidate, Akathisia → Cyproheptadine

Question **17**

Not answered

Marked out of 9.00

Flag

Management of drug-induced ECG changes

Mr. Y is a 55-year-old gentleman treated for psychosis and is on 600 mg of quetiapine. You are monitoring his cardiac functions by recording an ECG. What actions will be appropriate in the following situations?

Choose...
Reduce the dose and refer to a cardiologist immediately
Arrange for an echocardiogram
Prescribe diltiazem
Repeat ECG, stop the drug and refer to cardiologist immediately
Arrange for treadmill test
Reduce the dose and repeat ECG
Increase the dose of quetiapine
Alter the time of administration
No action required

QTc more than 500ms	Choose...
QTc less than 440ms	Choose...
QTc interval of 470 ms	Choose...
Abnormal T wave morphology	Choose...

Explanation:

Many psychotropic drugs are associated with ECG changes, and it is probable that certain drugs are causally linked to serious ventricular arrhythmias and sudden cardiac death. Therefore measure QTc in all patients prescribed antipsychotics. If QTc interval is less than 440 ms in men or less than 470 ms in women, no action is generally required unless abnormalities are noted in T-wave morphology in which case the treatment must be reviewed and urgent advice sought from the cardiology team. QTc of more than 440 ms in men or more than 470 ms in women but less than 500 ms must prompt consideration for reducing dose or switching to drug of lower effect; also repeat the ECG and consider referral to a cardiologist. QTc interval of more than 500 ms necessitates stopping the suspected causative drug and switching to a drug with lower QTc effect; this also requires a prompt referral to the cardiologist immediately.

QTc	Action	Refer to cardiologist
<440ms (men) or <470ms (women)	None unless abnormal T-wave morphology	Consider if in doubt
>440ms (men) or >470ms (women) but <500ms	Consider reducing dose or switching to drug of lower effect; repeat ECG	Consider
>500ms	Repeat ECG. Stop suspected causative drug(s) and switch to drug of lower effect	Immediately
Abnormal T-wave morphology	Review treatment. Consider reducing dose or switching to drug of lower effect	Immediately

SPMM COURSE

Adapted from Maudsley Prescribing Guidelines, 2021 Pg 145

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 145.

The correct answer is: QTc more than 500ms → Repeat ECG, stop the drug and refer to cardiologist immediately, QTc less than 440ms → No action required, QTc interval of 470 ms → Reduce the dose and repeat ECG, Abnormal T wave morphology → Reduce the dose and refer to a cardiologist immediately

Question **18**

Not answered

Marked out of 7.00

Remedial measures following rapid tranquillisation

Mr. Y is a 45-year-old gentleman admitted to the psychiatric intensive care unit for agitation and extremely disturbed behaviour. He was repeatedly given 1 mg of lorazepam and 5 mg of haloperidol intramuscularly. For each of the following problems, identify the important action/s to be taken.

Increased temperature (Choose ONE) — Choose...

Fall in blood pressure by more than 30 mm Hg — Choose...

Acute dystonia — Choose...

Irregular or slow pulse (less than 50/minute) — Choose...

Benzodiazepine induced respiratory depression — Choose...

Choose...
Give Flumazenil
Initiate CPR
Give Oxygen
Refer to specialist medical care immediately
Give procyclidine 5-10 mg IM
Ventilate mechanically on the psychiatric ward
Make the patient to lie flat
Ensure patient is not lying face-down
Attempt cardioversion
Check creatinine kinase urgently

Explanation:

Increased temperature following Rapid Tranquillisation (RT) must prompt a blood test for creatinine kinase urgently, due to a high risk of developing the neuroleptic malignant syndrome.

Fall in blood pressure during RT must be considered significant if more than 30 mm Hg orthostatic drop occurs or if the diastolic BP is less than 50 mm Hg. In this case, make the patient lie flat, tilt bed towards head and monitor closely.

Acute dystonia including oculogyric crisis can be managed by giving procyclidine 5-10mg IM or

benztropine 1-2 mg IM.

Irregular or slow pulse (less than 50/minute) is a sinister sign; it necessitates a referral to specialist medical care immediately.

Reduced respiratory rate of less than 10/min or oxygen saturation less than 90% must be managed by giving oxygen, raising the legs, ensuring patient is not lying face down, and giving flumazenil if benzodiazepine induced respiratory depression suspected. If any other sedative agent induces such a respiratory arrest, transfer to a medical bed immediately and ventilate mechanically.

Problem in rapid tranquillisation	Remedial measures
Acute dystonia (including oculogyric crises)	Give procyclidine 5-10 mg IM or IV
Reduced respiratory rate (<10/min) or oxygen saturation (<90%)	Give oxygen, raise legs, ensure patient is not lying face down Give flumazenil if benzodiazepine-induced respiratory depression suspected If induced by any other sedative agent: transfer to a medical bed and ventilate mechanically
Irregular or slow (<50/min) **pulse**	Refer to specialist medical care immediately
Fall in blood pressure (>30mmHg orthostatic drop or <50mmHg diastolic)	Have patient lie flat, tilt bed towards head Monitor closely
Increased temperature	Withhold antipsychotics (risk of NMS and perhaps arrhythmia) Check creatine kinase urgently

Adapted from Maudsley Prescribing Guidelines, 2021. pp. 63-4.

SPMM COURSE

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 63-64.

The correct answer is: Increased temperature (Choose ONE) → Check creatinine kinase urgently, Fall in blood pressure by more than 30 mm Hg → Make the patient to lie flat, Acute dystonia → Give procyclidine 5-10 mg IM, Irregular or slow pulse (less than 50/minute) → Refer to specialist medical care immediately, Benzodiazepine induced respiratory depression → Give Flumazenil

Question **19**

Not answered

Marked out of 7.00

Flag question

HiY Pharmacology CXEMI019

Drug Treatments

Select the drug that could be used, for the treatment of each of the following situations;

1. Catatonia
2. Neuroleptic malignant syndrome
3. Mild SIADH induced by antipsychotics
4. Antidepressant-induced hyponatraemia (2 answers)
5. Wernicke's encephalopathy

Benzodiazepines
Ropinirole
Dantrolene
Demeclocycline
Fluid Restriction
Venlafaxine
Clomipramine
Thiamine
Agomelatine
Ketamine
Selegiline

Explanation:

Benzodiazepines are the drugs of choice for catatonia occurring in the context of affective and conversion disorders. Catatonia in schizophrenia may be somewhat less likely to respond to benzodiazepines alone, with a response in the range of 40-50% of cases. The vast majority of evidence published recently suggests that prompt ECT remains the most successful treatment for catatonia.

Neuroleptic malignant syndrome: Symptomatic management for vital signs instability, fluid replacement and prevention of renal failure are the main treatment methods after immediate stopping of the causative psychotropic agent. **Treatment of neuroleptic malignant syndrome includes bromocriptine + dantrolene, sedation with benzodiazepines, and artificial ventilation if required.** Dantrolene is a peripheral muscle relaxant, which inhibits the intracellular calcium release from the sarcoplasmic reticulum. It was originally applied to treat cases of malignant hyperthermia.

If mild, SIADH induced by antipsychotics can be managed through fluid restriction with careful monitoring of serum sodium. A referral should be placed to specialist medical care if Na<125mmol/L. There are reports of demeclocycline use (interferes with ADH and increases water excretion).

Antidepressant-induced hyponatremia can be managed by water restriction, increased solute intake and prescribing non-SSRI agents, especially noradrenergic drugs such as reboxetine, lofepramine or an MAOI such as moclobemide. Demeclocycline may be useful in recurrent cases of both SIADH and hyponatremia.

Wernicke encephalopathy should be treated with high-potency parenteral vitamin B1 (thiamine) replacement. Avoid carbohydrate load until thiamine replacement is complete (i.e. do not rehydrate with glucose solutions prior to thiamine).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 132, 135-136, 165.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 606.

The correct answer is:

Catatonia- Benzodiazepines

Neuroleptic malignant syndrome- Dantrolene

Mild SIADH induced by antipsychotics- Fluid Restriction

Recurrent antidepressant-induced hyponatraemia- Demeclocycline, Fluid Restriction

Wernicke's encephalopathy- Thiamine

Question **20**

Not answered

Marked out of 5.00

Flag question

Receptor Pharmacology

Which of the listed terms best describes each of the following receptors that are relevant for pharmacological actions of various psychotropics?

Dopamine D2 receptor — Choose...

Glucocorticoid receptor — Choose...

Depolarization triggers activation in this receptor — Choose...

GABA-A receptor complex — Choose...

A receptor that regulates the synthesis of mediators other

Choose...
Inhibitory auto-receptor
Oligomer
Heteroreceptor
Heterodimer
Allosteric binding site
Nuclear receptor
Seven transmembrane receptor
Extrasynaptic receptor
Voltage-gated ion channel

than its own ligand — Choose...

Explanation:

Receptors linked to G proteins, including the dopamine D2 receptor, all have the structure of seven-transmembrane regions, meaning that they span the membrane seven times.

Glucocorticoids (GCs) are steroid hormones widely used for the treatment of inflammation, autoimmune diseases, and cancer. To exert their broad physiological and therapeutic effects, GCs bind to the GC receptor which belongs to the nuclear receptor subfamily of transcription factors.

The opening and closing of voltage-sensitive or voltage-gated ion channels are regulated by the ionic charge or voltage potential across the membrane in which they reside. An electrical impulse in a neuron, also known as the action potential, is triggered by the summation of various neurochemical and electrical events of neurotransmission.

Benzodiazepine-sensitive GABA-A receptors consist of five subunits with a central chloride channel and have binding sites not only for GABA but also for positive allosteric modulators (e.g. benzodiazepines). When a positive allosteric modulator such as a benzodiazepine binds to the GABA-A receptor in the presence of GABA, it causes the channel to open even more frequently than when GABA alone is present.

When a receptor for a neurotransmitter other than the one the neuron uses as its own neurotransmitter is present, it is called a 'heteroreceptor' (literally, other receptor).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 36, 66-67, 125, 262.

Timmermans S et al. A general introduction to glucocorticoid biology. Frontiers in Immunology. 2019 Jul 4;10:1545.

The correct answer is: Dopamine D2 receptor → Seven transmembrane receptor, Glucocorticoid receptor → Nuclear receptor, Depolarization triggers activation in this receptor → Voltage-gated ion channel, GABA-A receptor complex → Allosteric binding site, A receptor that regulates the synthesis of mediators other than its own ligand → Heteroreceptor

Question **21**

Not answered

Marked out of 5.00

Newer antidepressants

Select mechanism of action for the following drugs

Antidepressant effects of mirtazapine — Choose...

Antidepressant effects of agomelatine — Choose...

Antidepressant effects of duloxetine — Choose...

Oral selegiline — Choose...

Choose...
MAOB-B inhibitor
Presynaptic alpha-2 agonist
MAOI-A inhibitor
5-HT2A receptor stimulation
D2 receptor partial agonist
Alpha-2 adrenoreceptor and 5-HT2C antagonism
Norepinephrine and serotonin reuptake reduction
NMDA receptor antagonism
GABA-A Agonist
5-HT2C receptor antagonism

Explanation:

Mirtazapine is a multifunctional drug with five principal mechanisms of action: 5HT-2A, 5HT-2C, 5HT-3, alpha-2 adrenergic, and H1 histamine antagonism. Of these, its alpha 2 antagonism and 5HT-2C blockade are seen as primary mechanisms of antidepressant function. It also displays antihistaminic and anti-5HT3 properties. Mianserin has similar profile except for antihistaminic properties, but it has more anticholinergic properties than mirtazapine.

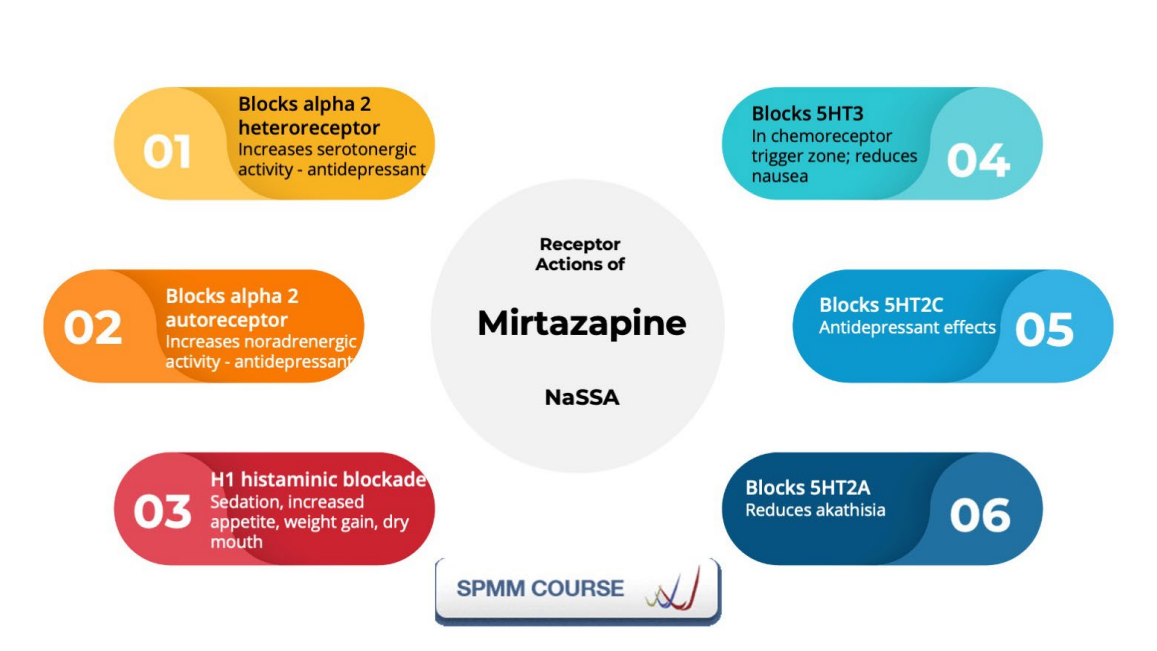
Agomelatine enhances norepinephrine and dopamine neurotransmission through 5-HT2C antagonism. It is also a direct agonist at melatonin (MT1 and MT2) receptors. GABA interneurons tonically inhibit noradrenergic circuits (from locus coeruleus) and dopaminergic circuits (from ventral tegmentum) projecting to prefrontal cortex. Serotonin via 5HT2C stimulation drives these GABA interneurons. Thus, norepinephrine and dopamine circuits are inhibited by the normal tonic release of serotonin onto 5-HT2C receptors (Stahl, 2007). Thus agomelatine, through 5HT2C inhibition, acts as norepinephrine and dopamine disinhibitor (NDDI).

Serotonin-norepinephrine reuptake inhibitors (SNRIs) combine robust serotonin transporter (SERT) inhibition with varying degrees of inhibition of the norepinephrine transporter (NET). Duloxetine is an SNRI that is characterised pharmacologically by slightly more potent SERT than NET inhibition.

Oral selegiline is a Monoamine Oxidase Inhibitor that is selective for MAO-B at normal therapeutic doses; this selectivity is lost when a transdermal selegiline patch is applied at higher doses, leading to notable antidepressant action. In the transdermal form, selegiline is a tissue-selective MAO inhibitor (MAO-A and MAO-B inhibitor in the brain and relatively selective MAO-B inhibitor in the gut).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 298-302, 306-309.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 705.



The correct answer is: Antidepressant effects of mirtazapine
→ Alpha-2 adrenoreceptor and 5-HT2C antagonism, Antidepressant effects of agomelatine → 5-HT2C receptor antagonism, Antidepressant effects of duloxetine → Norepinephrine and serotonin reuptake reduction, Oral selegiline → MAOB-B inhibitor

Question **22**

Not answered

Marked out of 5.00

Flag question

Effect of receptor-level actions (1)

Match the receptors below with its most significant effect from the list shown here.

Blockade of D-2 receptors in the mesolimbic pathway — Choose...

Blockade of D-2 receptors in the mesocortical pathway — Choose...

Choose...
Production of negative symptoms
Insomnia and sexual side effects of SSRIs
Dry mouth and constipation
Therapeutic effects of antidepressants
Gastrointestinal side effects of SSRIs
Leads to tardive dyskinesia
Leads to features of hyperprolactinaemia
Leads to extrapyramidal side effects
Sedation and orthostatic hypotension
Sedation and weight gain
Therapeutic effects of antipsychotics

Blockade of D-2 receptors in the nigrostriatal pathway	Choose...
Blockade of D-2 receptors in the tuberoinfundibular pathway	Choose...
Dopamine receptor supersensitivity	Choose...

Explanation:

By the 1970s it was widely recognised that the key pharmacological property of all 'neuroleptics' with antipsychotic properties was their ability to block D2 receptors, specifically those in the mesolimbic/mesostriatal dopamine pathway.

The mesocortical dopamine (DA) pathway arises from cell bodies in the ventral tegmental area, projecting to areas of the prefrontal cortex. Hypoactivity of dopamine neurons in the mesocortical dopamine pathway theoretically mediates the cognitive, negative, and affective symptoms of schizophrenia.

The nigrostriatal dopamine pathway projects from the substantia nigra to the basal ganglia or striatum. It is part of the extrapyramidal nervous system and plays a key role in regulating movements. When dopamine is deficient, it can cause parkinsonism with tremor, rigidity, and akinesia/bradykinesia.

The dopamine neurons that project from hypothalamus to anterior pituitary gland are known as the tuberoinfundibular dopamine pathway. Normally, these neurons are tonically active and inhibit prolactin release. If the functioning of the tuberoinfundibular dopamine neurons is disrupted by lesions or drugs, prolactin levels can rise.

One theory explaining tardive dyskinesia (TD) is that nigrostriatal D2 receptors most sensitive to blockade trigger a form of undesirable neuroplasticity called super-sensitivity in reaction to D2 receptor blockade.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 85-87, 95, 162, 170.

The correct answer is: Blockade of D-2 receptors in the mesolimbic pathway → Therapeutic effects of antipsychotics, Blockade of D-2 receptors in the mesocortical pathway → Production of negative symptoms, Blockade of D-2 receptors in the nigrostriatal pathway → Leads to extrapyramidal side effects, Blockade of D-2 receptors in the tuberoinfundibular pathway → Leads to features of hyperprolactinaemia, Dopamine receptor supersensitivity → Leads to tardive dyskinesia

Question **23**

Not answered

Marked out of 5.00

Effect of receptor-level actions (2)

Match the receptor below with its most significant effect from the list above

Choose...
Tolerance to hallucinogenic drugs
Myocarditis
Sedation and weight gain
Dry mouth, constipation
Leads to tardive dyskinesia
Production of negative symptoms
Therapeutic effects of antidepressants
Sedation and orthostatic hypotension
QT prolongation
Neuroleptic malignant syndrome
Insomnia and sexual side effects of SSRIs

Blockade of postsynaptic alpha-1 adrenoreceptor	Choose...
Blockade of M3 cholinergic receptors	Choose...
Blockade of H1 histamine receptors	Choose...
Downregulation of 5-HT2A receptors	Choose...
D-2 receptor blockade	Choose...

Explanation:

Alpha-1 adrenergic blockade can cause dizziness, sedation, and hypotension.

Side effects of muscarinic cholinergic receptor blockade include dry mouth, blurred vision, urinary retention, and constipation, as well as central side effects including drowsiness and cognitive dysfunction (such as problems with memory, concentration, and slowing of cognitive processing).

It is widely appreciated that antihistamines are sedating. When histamine binds to postsynaptic histamine-1 (H1) receptors, it activates a G-protein-linked second-messenger system that activates phosphatidyl inositol and the transcription factor cFOS. This results in wakefulness and normal alertness. H1 antagonists prevent activation of this second messenger and thus can cause sleepiness.

Hallucinogens can produce incredible tolerance, sometimes after a single dose. Desensitisation of 5HT-2A receptors is hypothesised to underlie this rapid clinical and pharmacological tolerance.

A rare but potentially fatal complication can occur with D2 receptor blockade, possibly due in part to D2 receptor blockade specifically in the nigrostriatal motor pathway. This is called the 'neuroleptic malignant syndrome', associated with extreme muscular rigidity, high fevers, coma, and even death.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 169, 425-427, 568.

The correct answer is: Blockade of postsynaptic alpha-1 adrenoreceptor → Sedation and orthostatic hypotension, Blockade of M3 cholinergic receptors → Dry mouth, constipation, Blockade of H1 histamine receptors → Sedation and weight gain, Downregulation of 5-HT2A receptors → Tolerance to hallucinogenic drugs, D-2 receptor blockade → Neuroleptic malignant syndrome

Question **24**

Not answered

Marked out of 3.00

Side effects and receptors

Choose one explanatory option for each of the following conditions:

Clozapine induced hypersalivation	Choose...
SSRI-induced nausea	Choose...
Lower incidence of EPSEs of newer atypical antipsychotics	Choose...

Choose...
Low D1 receptor sensitivity
High 5HT2 Serotonin receptor sensitivity
M4 agonism
Dopamine D4 activation
Serotonin 5HT3 activation
Decreased nicotinic receptor density
Apolipoprotein E receptors
Serotonin 5HT2 receptor blockade
Serotonin 5HT3 blockade
Low D3 receptor occupancy in striatum

Explanation:

The pharmacological basis of clozapine-related hypersalivation remains unclear. Suggested mechanisms include muscarinic M4 agonism, adrenergic alpha-2 antagonism, and inhibition of the swallowing reflex.

5HT3 receptors are localised in the chemoreceptor trigger zone of the brainstem, where they mediate nausea and vomiting (especially in response to cancer chemotherapy), and in the gastrointestinal tract itself where they mediate nausea, vomiting, and diarrhea/bowel motility when stimulated by serotonin (including when stimulated by serotonin that is a side effect of peripherally increased serotonin by SSRIs/SNRIs). Blocking these 5HT3 receptors can therefore protect against chemotherapy-induced nausea and vomiting as well as against serotonin-induced gastrointestinal side effects that can accompany agents that increase serotonin.

Lower incidence of EPSEs of newer atypical antipsychotics is due to 5-HT2 receptor blockade. PET and SPECT studies have shown that newer atypicals like olanzapine, risperidone, quetiapine and clozapine have a higher degree of 5-HT2A receptor occupancy

(more than 90%) that leads to a reduced incidence of EPSEs.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 227.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 309-311.

The correct answer is: Clozapine induced hypersalivation → M4 agonism, SSRI-induced nausea → Serotonin 5HT3 activation, Lower incidence of EPSEs of newer atypical antipsychotics → Serotonin 5HT2 receptor blockade

Question **25**

Not answered

Marked out of 5.00

Choose...
Gabapentin
Vigabatrin
Lithium
Lamotrigine
Topiramate
Levetiracetam
Pregabalin
Sodium Valproate
Carbamazepine

Mechanism of action

Identify the correct drug based on their mechanism of action

Enhances GABAergic function and inhibits GABA aminotransferase	Choose...
Effects on intracellular second messenger systems	Choose...
Neuronal membrane stabiliser and blocks voltage-dependent sodium channels	Choose...
Decrease GABA metabolism by inhibition of glial GABA transaminase	Choose...
Enhances GABA and blocks glutamate at NMDA receptors	Choose...

Explanation:

As for all anticonvulsants, the exact mechanism of action of valproic acid is uncertain; however, even less may be known about the mechanism of valproate than for other anticonvulsants. At least three possibilities exist for how valproic acid works: inhibiting voltage-sensitive sodium channels, boosting the actions of the neurotransmitter GABA, and regulating downstream signal transduction cascades. Sodium valproate also inhibits GABA aminotransferase, and increases GABA binding in some brain structures like the hippocampus.

Lithium is an ion whose mechanism of action is not certain. Candidates for its mechanism of actions are various signal transduction sites beyond neurotransmitter receptors. This includes second messengers such as the phosphatidyl inositol system, where lithium inhibits the enzyme inositol monophosphatase; modulation of G proteins; and, most recently, regulation of gene expression for growth factors and neuronal plasticity by interaction with downstream signal transduction cascades, including inhibition of GSK-3 and protein kinase C.

Lamotrigine may work by blocking the alpha subunit of voltage-sensitive sodium channels and could perhaps also have actions at other ion channels for calcium and potassium. Lamotrigine thus stabilises neuronal membranes. It is also thought to reduce the release of the excitatory neurotransmitter glutamate.

Vigabatrin is an irreversible inhibitor of GABA-transaminase (GABA-T), an enzyme that degrades GABA. Apart from inhibiting GABA-T, vigabatrin prevents neuronal uptake of GABA and stimulates its release into the synapse. (**Vi GABA TR**ansaminase **IN**hibitor).

Topiramate blocks voltage-sensitive sodium channels by an unknown mechanism, inhibits the release of glutamate, potentiates activity of GABA, and is a carbonic anhydrase inhibitor.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 345-347, 351.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 781.

The correct answer is: Enhances GABAergic function and inhibits GABA aminotransferase → Sodium Valproate, Effects on intracellular second messenger systems → Lithium, Neuronal membrane stabiliser and blocks voltage-dependent sodium channels → Lamotrigine, Decrease GABA metabolism by inhibition of glial GABA transaminase → Vigabatrin, Enhances GABA and blocks glutamate at NMDA receptors → Topiramate

Question **26**

Not answered

Marked out of 4.00

Management of mood disorder

Choose the best treatment option from the shown list for each of the following clinical situations

Rob is a 40-year-old man with bipolar disorder currently depressed. He has been on lithium for five years. Choose...

Lucy is a 25-year-old woman who has a history of bipolar disorder. She is now presenting with a mixed affective state. She is also pregnant for the first time. Choose...

Andy is a 50-year-old man who suffers from recurrent depression treated by his GP with venlafaxine. He has not made much improvement. While he has a partial response to venlafaxine, switching antidepressants has previously been unsuccessful. Choose...

Linda is a 35-year-old woman who has a history of recurrent depressive episodes followed by hypomania often due to her antidepressant drugs. She recovers from the most recent episode of depression. Her GP is keen to start her on prophylactic treatment. Choose...

Choose...
Lamotrigine
Haloperidol
Citalopram
Sodium valproate
Mirtazapine
Lithium
Carbamazepine
Chlorpromazine
Venlafaxine

As reflected in the most recent bipolar treatment guidelines, if the patient is already taking lithium, the level should be checked and optimised. If lithium is optimised, there are several options for the treatment of bipolar depression, including the addition of olanzapine+fluoxetine, quetiapine, olanzapine, or lamotrigine.

Because antipsychotics have a better risk/benefit ratio than valproate and carbamazepine, and a larger reproductive evidence base and possibly better reproductive safety profile than lamotrigine and lithium, they generally form the mainstay of treatment for women in the perinatal period. No mood stabiliser is clearly safe. NICE recommends the use of mood-stabilising antipsychotics as a preferable alternative to a mood stabiliser. SGAs generally have a larger evidence base than FGAs for use in bipolar disorder. The most reproductive safety data are available for quetiapine, olanzapine, risperidone, and haloperidol.

Current guidelines for the treatment of depression recommend considering adding a second agent especially if there is partial response on the current antidepressant and there is good tolerability of the current antidepressant and switching antidepressants has been unsuccessful. It is recommended to consider adding quetiapine, aripiprazole, or lithium as first-line treatments.

Lithium is considered first-line long-term treatment for bipolar disorder. The International Society for Bipolar Disorders consensus on the use of antidepressants in bipolar patients has highlighted the clinical consensus discouraging their use in patients with rapid cycling, depressive episodes with mixed features, and as monotherapy.

Ref: National Institute for Health and Clinical Excellence. The NICE guideline on the assessment and management of bipolar disorder in adults, children and young people in primary and secondary care. Updated edition 2020.

Goodwin GM et al. Evidence-based guidelines for treating bipolar disorder: revised third edition recommendations from the British Association for Psychopharmacology. Journal of Psychopharmacology. 2016 Jun;30(6):495-553.

McAllister-Williams RH et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. Journal of Psychopharmacology. 2017 May;31(5):519-52.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021.

Cleare A et al. Evidence-based guidelines for treating depressive disorders with antidepressants: a revision of the 2008 British Association for Psychopharmacology guidelines. Journal of Psychopharmacology. 2015 May;29(5):459-525.

The correct answer is: Rob is a 40-year-old man with bipolar disorder currently depressed. He has been on lithium for five years. → Lamotrigine, Lucy is a 25-year-old woman who has a history of bipolar disorder. She is now presenting with a mixed affective state. She is also pregnant for the first time. → Haloperidol, Andy is a 50-year-old man who suffers from recurrent depression treated by his GP with venlafaxine. He has not made much improvement. While he has a partial response to venlafaxine, switching antidepressants has previously been unsuccessful. → Lithium, Linda is a 35-year-old woman who has a history of recurrent depressive episodes followed by hypomania often due to her antidepressant drugs. She recovers from the most recent episode of depression. Her GP is keen to start her on prophylactic treatment. → Lithium

Question **27**

Not answered

Marked out of 6.00

Flag question

Pharmacological properties

Identify the correct drug for each of the following descriptions

An antidepressant with high anticholinergic effect — Choose...

Excreted predominantly via renal route — Choose...

A drug with pronounced antihistaminic effect — Choose...

Choose...
Carbamazepine
Venlafaxine
Reboxetine
Gabapentin
Duloxetine
Sodium valproate
Clomipramine

Explanation:

Clomipramine can cause a number of side effects. Its anticholinergic activity may explain sedative effects, dry mouth, constipation, and blurred vision. Sedative effects and weight gain may be due to antihistamine properties. Its blockade of alpha adrenergic 1 receptors may explain dizziness, sedation, and hypotension. Cardiac arrythmias and seizures, especially in overdose, may be caused by blockade of ion channels.

Gabapentin is not metabolised but excreted intact renally.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 16 333.

The correct answer is: An antidepressant with high anticholinergic effect → Clomipramine, Excreted predominantly via renal route → Gabapentin, A drug with pronounced antihistaminic effect → Clomipramine

Question **28**

Not answered

Marked out of 5.00

Flag question

History of psychiatric treatments

Match each of the following biological treatments with the pioneers associated from the given list

Electroconvulsive therapy — Choose...

Lithium — Choose...

Choose...
Manfred Sakel
Ganesh Sen and Kartik Bose
John Cade
Hugo Sternbach
Janssen
Ugo Cereletti and Lucio Bini
F M Berger
Kuhn
Delay and Deniker
Egas Moniz and Almeida Lima

	Choose...
Chlorpromazine	Choose...
Haloperidol	Choose...
Chlordiazepoxide	Choose...

Explanation:

Hungarian neuropsychiatrist Ladislas von Meduna used camphor as an agent for the therapeutic induction of seizures. Two Italian psychiatrists, **Ugo Cerletti and Lucio Bini** were interested in the use of electricity to induce seizures. In 1938, they conducted the first electrical induction of a series of seizures in a catatonic patient and produced a successful treatment response.

The use of lithium in psychiatry goes back to the mid-19 th century. Early work, however, was soon forgotten, and **John Cade** is credited with reintroducing lithium to psychiatry for mania in 1949.

Early in their Vietnam War (late 1940s and early 1950s), French pharmacologists employed a combination of the antihistamine phenergan and an opioid medication in the wounded, facilitating analgesia, alleviating fear, and opposing shock. While working on similar compounds, **Delay and Deniker** developed the first antipsychotic agent,chlorpromazine, at the Rhône-Poulenc Pharmaceutical Company, under the direction of Paul Charpentier. First studied by a French military surgeon for trauma, psychiatrists soon adapted it for its 'tranquilising' effects.

In 1958, **Janssen** developed haloperidol, the first butyrophenone neuroleptic.

In 1955, chlordiazepoxide, the first benzodiazepine, was synthesised by **Sternbach** for Roche.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 729-730.

Shorter E. The history of lithium therapy. Bipolar disorders. 2009 Jun;11:4-9.

Westermeyer J. Cultural Psychiatry in the French-Speaking World. In Cultural Psychiatry 2013 (Vol. 33, pp. 56-63). Karger Publishers.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 38.

The correct answer is: Electroconvulsive therapy → Ugo Cereletti and Lucio Bini, Lithium → John Cade, Chlorpromazine → Delay and Deniker, Haloperidol → Janssen, Chlordiazepoxide → Hugo Sternbach

Question **29**

Not answered

Marked out of

Side effects of medication

Match the side effects with the drugs most commonly associated with it

Choose...
Sodium valproate
Agomelatine
Trazodone
Citalopram
Chlorpromazine
Lithium
Reboxetine
Thioridazine
Clozapine
Risperidone

Thrombocytopenia	Choose...
Gastric bleeding	Choose...
Hypothyroidism	Choose...
Cardiomyopathy	Choose...
Urinary hesitancy	Choose...

Explanation:

Mood stabilisers associated with thrombocytopaenia include carbamazepine, phenytoin, and valproic acid. The use of sodium valproate is associated with thrombocytopenia, leucopenia, and red cell hypoplasia. Blood counts including platelets should also be periodically monitored on valproate. Thrombocytopenia is a dose-related effect of valproate; a decrease in dose is required if bruising or bleeding gums is noted.

Serotonin is released from platelets in response to vascular injury, promoting vasoconstriction and morphological changes in platelets that lead to aggregation. SSRIs inhibit the serotonin transporter, which is responsible for the uptake of serotonin into platelets. This depletion of platelet serotonin leads to a reduced ability to form clots and a subsequent increase in the risk of bleeding. SSRIs may also increase gastric acid secretion and therefore may be indirectly irritant to the gastric mucosa. The use of serotonergic antidepressants, such as citalopram, is an independent risk factor for bleeding events.

Use of lithium is associated with an increased risk of developing hypothyroidism, especially in younger women. 5% patients may develop goiter and overt hyperthyroidism is also reported in some cases. Thyroid deficiency is common in those with high risk for preexisting antithyroid antibodies such as middle-aged women. The risk is 3-4:1 in women and is high in first two years of treatment. Rapid cycling patients are at higher risk. High TSH is seen in nearly 1/3rd of chronic lithium-treated patients - without clinical hypothyroidism. In resistant depression or non-responsive cyclers with bipolar it may be useful treating this subclinical state with thyroxine.

Clozapine is associated with cardiomyopathy. It may occur at any time. Cardiomyopathy should be suspected in any patient showing signs of heart failure, which should provoke immediate cessation of clozapine. Myocarditis seems to occur within 6-8 weeks of starting clozapine and cardiomyopathy may occur later in treatment (median-9 months) but both may occur at any time.

Reboxetine is a selective norepinephrine reuptake inhibitor. Side effects may appear ‘anticholinergic’, but reboxetine does not directly block muscarinic receptors. Constipation, dry mouth, and urinary retention are noradrenergic, due in part to peripheral alpha 1 receptor stimulation causing decreased acetylcholine release. Thus, antidotes for these side effects can be alpha 1 antagonists such as tamsulosin or doxazosin, especially in men over 50 with borderline urine flow. Urinary hesitancy has been observed in around 10% of male patients taking part in the clinical trials of reboxetine.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 222-223, 409-410.

Shine B, McKnight RF, Leaver L, Geddes JR. Long-term effects of lithium on renal, thyroid, and parathyroid function: a retrospective analysis of laboratory data. The Lancet. 2015 Aug 1;386(9992):461-8.

Stahl SM. Stahl’s Essential Psychopharmacology: Prescriber’s Guide, 7th Edition. 2021. pp. 681-68

The correct answer is: Thrombocytopenia → Sodium valproate, Gastric bleeding → Citalopram, Hypothyroidism → Lithium, Cardiomyopathy → Clozapine, Urinary hesitancy → Reboxetine

Question **30**

Classification of psychotropic drugs

Match each chemical class below with the most appropriate drug from the list.

Pyrazolopyrimidine	Choose...
Non hydrazine MAOI	Choose...
Reversible MAOI	Choose...
Full agonist at the BDZ receptor	Choose...

Choose...
Phenelzine
Zaleplon
Phentolamine
Isocarboxazid
Moclobemide
Zopiclone
Zolpidem
Clonazepam
Tranylcypromine
Diazepam

Partial agonist at the BDZ receptor — Choose...

Explanation:

Cyclopyrrolone - Zopiclone, Imidazopyridine -Zolpidem. Pyrazolopyrimidine- Zaleplon.

Monoamine oxidase inhibitors include both hydrazine and non-hydrazine derivatives. Hydrazine derivatives include phenelzine and isocarboxazid (greater hepatotoxicity than tranylcypromine, a non hydrazine compound). Moclobemide is a RIMA (a reversible inhibitor of monoamine oxidase A).

Among benzodiazepines, there are full agonists like diazepam and partial agonists like clonazepam, which act at the BDZ receptor to enhance the action of GABA.

The correct answer is: Pyrazolopyrimidine → Zaleplon, Non hydrazine MAOI → Tranylcypromine, Reversible MAOI → Moclobemide, Full agonist at the BDZ receptor → Diazepam, Partial agonist at the BDZ receptor → Clonazepam

Question **31**

Not answered

Marked out of 5.00

Antidepressant side effects

Choose the most likely offending drug for each of the following situations.

A 61-year-old lady develops dry mouth, blurred vision, and confusion v... taking this medication for depression. — Choose...

A 44-year-old man develops septicaemia, and an FBC shows agranulocytosis following anti-depressant treatment. — Choose...

An 11-year-old boy develops sudden involuntary jerky movements in his face with vocal grunts. — Choose...

A 64-year-old woman is prescribed amitriptyline for many years by her GP. She has a relapse in depression, and the GP changes her prescription to another tricyclic agent. But she takes three weeks supply of this antidepressant as an overdose. She develops arrhythmias and despite intensive cardiac monitoring, dies three days later. — Choose...

A 62-year-old man is on treatment for intractable OCD. He develops myoclonus, nausea, diarrhoea and disinhibition when citalopram is added to his first medication. — Choose...

Choose...
Venlafaxine
Fluoxetine
Mirtazapine
Amitriptyline
Clomipramine
Imipramine
Methylphenidate
Moclobemide
Dosulepin
Esketamine
Phenelzine
Lofepramine

Explanation:

Amitriptyline is a tricyclic antidepressant that acts as a serotonin and norepinephrine reuptake inhibitor. Its blockade of alpha adrenergic 1 receptors can cause dizziness, sedation, and hypotension. Sedative effects and weight gain can occur due to antihistaminic properties. Anticholinergic activity can cause sedative effects, dry mouth, constipation, blurred vision, and confusion.

Mirtazapine has been associated with leucopenia and, rarely, agranulocytosis.

Drugs-induced tics have been described with psychostimulants, antipsychotics, and anti-epileptic drugs. Methylphenidate has traditionally been associated with the appearance of tics due to the increased dopamine activity caused by stimulants. However, several recent studies have also highlighted the fact that tics can improve with the use of methylphenidate in patients suffering from ADHD associated with tic disorders.

Dothiepin (dosulepin) and amitriptyline are the most toxic tricyclics in overdose (seizures and cardiac arrythmias).

Serotonin syndrome, a potentially life-threatening acute reaction to serotonergic drugs, occurs more frequently with a combination of serotonergic drugs or in overdose.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 26.

Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. pp. 149, 683.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 290.

Touafchia D et al. Drug-induced tics: An observational postmarketing study. Human Psychopharmacology: Clinical and Experimental. 2020 Jul;35(4):e2734.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 674.

The correct answer is: A 61-year-old lady develops dry mouth, blurred vision, and confusion when taking this medication for depression. → Amitriptyline, A 44-year-old man develops septicaemia, and an FBC shows agranulocytosis following anti-depressant treatment. → Mirtazapine, An 11-year-old boy develops sudden involuntary jerky movements in his face with vocal grunts. → Methylphenidate, A 64-year-old woman is prescribed amitriptyline for many years by her GP. She has a relapse in depression, and the GP changes her prescription to another tricyclic agent. But she takes three weeks supply of this antidepressant as an overdose. She develops arrhythmias and despite intensive cardiac monitoring, dies three days later. → Dosulepin, A 62-year-old man is on treatment for intractable OCD. He develops myoclonus, nausea, diarrhoea and disinhibition when citalopram is added to his first medication. → Clomipramine

Question **32**

Not answered

Marked out of 4.00

Flag

Receptor mechanisms

Identify the drugs & receptors that display the following features:

Non-competitive antagonist: Choose...

Irreversible antagonists: Choose...

Full agonists: Choose...

Choose...
Buspirone
Phencyclidine at NMDA receptors
MAOIs
Atropine at muscarinic receptors
Buprenorphine
Aripiprazole
Propranalol at beta adrenergic receptors
Benzodiazepines at GABA A complex

Explanation:

Phencyclidine (PCP) blocks NMDA receptors as a non-competitive antagonist at a site inside the ion channel. The anaesthetic ketamine binds as an antagonist to the open channel conformation of the NMDA receptor. Specifically, it binds to a site within the calcium channel of this receptor, which is often termed the PCP site because it is also where PCP binds as an antagonist.

The MAOIs phenelzine, tranylcypromine, isocarboxazid, and selegiline are all irreversible enzyme inhibitors, and thus enzyme activity returns only after new enzyme is synthesised about 2-3 weeks later.

Benzodiazepines belong to a class of drugs that modulate the neurotransmitter GABA, especially GABA-A, to open the chloride channel. Benzodiazepines have a powerfully calming effect on the brain. They modulate GABA-A receptors as full agonists.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 106-110, 337.

Fluyau D et al. Challenges of the pharmacological management of benzodiazepine withdrawal, dependence, and discontinuation. Therapeutic advances in psychopharmacology. 2018 May;8(5):147-68.

The correct answer is: Non-competitive antagonist → Phencyclidine at NMDA receptors, Irreversible

antagonists → MAOIs, Full agonists → Benzodiazepines at GABA A complex

Question **33**

Not answered

Marked out of 6.00

Swapping antidepressants

For the clinical situations given below, choose the best management option.

A 45-year-old woman on paroxetine for four weeks has now decided to take citalopram, as she responded well to this medication in the past. Choose...

A 57-year-old woman on fluoxetine has atypical depression. Now you have decided to treat her on moclobemide. Choose...

A 46-year-old man with depression on tranylcypromine has decided to switch over to sertraline Choose...

A 52-year-old woman on Venlafaxine developed insomnia & dizzy spells. She is not keen to continue taking this antidepressant. She is now willing to take mirtazapine. Choose...

Choose...
It may cause withdrawal symptoms
Withdraw and then start new drug straightaway
Withdraw and wait two weeks before starting new drug
Cross taper cautiously
Co-administer for a brief period
Withdraw and wait one week before starting new drug
Direct switch is possible
Stop drug immediately
Discontinue over a month
Withdraw and wait five weeks before starting new drug

Explanation:

When switching from one SSRI to another, their effects are so similar that administration of the second drug is likely to ameliorate withdrawal effects of the first. Therefore cross-tapering is not necessary, and a direct switch is possible.

When switching from fluoxetine to moclobemide, stop fluoxetine then wait 5-6 weeks before starting moclobemide.

When switching from tranylcypromine to sertraline, tranylcypromine should be tapered and stopped; then wait 2 weeks before starting the SSRI.

When switching from venlafaxine to mirtazapine, cross-taper cautiously.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 336-341.

The correct answer is: A 45-year-old woman on paroxetine for four weeks has now decided to take citalopram, as she responded well to this medication in the past. → Direct switch is possible, A 57-year-old woman on fluoxetine has atypical depression. Now you have decided to treat her on moclobemide. → Withdraw and wait five weeks before starting new drug, A 46-year-old man with depression on tranylcypromine has decided to switch over to sertraline → Withdraw and wait two weeks before starting new drug, A 52-year-old woman on Venlafaxine developed insomnia & dizzy spells. She is not keen to continue taking this antidepressant. She is now willing to take mirtazapine. → Cross taper cautiously

Question **34**

Not answered

Marked out of 5.00

Drug action on neurotransmitters

For each of the examples given below, identify the level of drug action on neurotransmitters;

The suspected mechanism of therapeutic action of lithium via inositol	Choose...
Benzodiazepines at GABA-A complex	Choose...
High-dose amphetamine abuse on dopamine and noradrenaline	Choose...
Therapeutic mechanism of monoamine oxidase inhibitors	Choose...
Lofexidine at alpha-2 receptors	Choose...

Choose...
Pre-synaptic receptor activity
Antagonists
Storage of neurotransmitter
Synthesis of neurotransmitter
Irreversible enzyme inhibition
Release from storage
Second messengers
Postsynaptic receptors
Full agonist
Re-uptake
Partial agonist

Explanation:

Lithium is an ion whose mechanism of action is not certain. Candidates for its mechanism of actions are various signal transduction sites beyond neurotransmitter receptors. This includes second messengers such as the phosphatidyl inositol system, where lithium inhibits the enzyme inositol monophosphatase; modulation of G proteins; and, most recently, regulation of gene expression for growth factors and neuronal plasticity by interaction with downstream signal transduction cascades, including inhibition of GSK-3 and protein kinase C.

Benzodiazepines belong to a class of drugs that modulate the neurotransmitter GABA, especially GABA-A, to open the chloride channel. Benzodiazepines have a powerfully calming effect on the brain. They modulate GABA-A receptors as full agonists.

Oral administration of clinically approved doses of the stimulant amphetamine block the transporters for both norepinephrine (NE) and dopamine (DA) (NETs and DATs). Thus, DA and NE reuptake are inhibited. At the high doses of amphetamine used by stimulant addicts, additional actions are triggered. Following competitive inhibition of DATs, amphetamine is transported as a hitchhiker into the presynaptic DA terminal. Once there in sufficient quantities, amphetamine is also a competitive inhibitor of the vesicular transporter (VMAT2) for both DA and NE. Once amphetamine hitch-hikes another ride into synaptic vesicles, it displaces DA there, causing a flood of DA release. As DA accumulates in the cytoplasm of the presynaptic neuron, it causes the DATs to reverse directions, spilling intracellular DA into the synapse, and also opening presynaptic channels to further release DA in a flood into the synapse.

The MAOIs phenelzine, tranylcypromine, isocarboxazid, and selegiline are all irreversible enzyme inhibitors, and thus enzyme activity returns only after new enzyme is synthesised about 2-3 weeks later.

Lofexidine is a non-opioid medication that acts as an agonist at central alpha-2-adrenergic presynaptic receptors and thereby suppresses noradrenergic hyperactivity.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 337, 345-346, 472.

Fluyau D et al. Challenges of the pharmacological management of benzodiazepine withdrawal, dependence, and discontinuation. Therapeutic advances in psychopharmacology. 2018 May;8(5):147-68.

Alam D et al. Efficacy of lofexidine for mitigating opioid withdrawal symptoms: results from two randomized, placebo-controlled trials. Journal of drug assessment. 2020 Jan 1;9(1):13-9.

The correct answer is: The suspected mechanism of therapeutic action of lithium via inositol → Second messengers, Benzodiazepines at GABA-A complex → Full agonist, High-dose amphetamine abuse on dopamine and noradrenaline → Release from storage, Therapeutic mechanism of monoamine oxidase inhibitors → Irreversible enzyme inhibition, Lofexidine at alpha-2 receptors → Pre-synaptic

receptor activity

Question **35**

Not answered

Marked out of 4.00

Common adverse effects

For the drugs given below identify the most common side effects from the list.

Lithium — Choose...

Venlafaxine — Choose...

Carbamazepine in the elderly — Choose...

Choose...
Diplopia
Postural hypotension
Insomnia
Hyponatraemia
Polyuria
Sweating
Hair loss
Agranulocytosis
Ataxia
Weight gain

Explanation:

Polyuria due to impaired urinary concentrating ability occurs in up to 20% of patients chronically treated with lithium. Adverse effects of lithium also include gastrointestinal upset and fine tremor.

Depending on the dose, venlafaxine has different degrees of inhibition of serotonin reuptake (most potent and robust even at low doses) versus norepinephrine reuptake (moderate potency and robust only at higher doses). Adding norepinephrine transporter inhibition likely accounts for two side effects of venlafaxine in some patients: sweating and elevated blood pressure. Venlafaxine is available as an extended-release formulation, which not only allows for once-daily administration but also significantly reduces side effects, especially nausea. In contrast to several other psychotropic drugs available in controlled-release formulations, extended-release venlafaxine is a considerable improvement over the immediate-release formulation, which has fallen into little or no use because of unacceptable nausea and other side effects, especially when immediate-release venlafaxine is started or when it is stopped.

Antidiuretic effects leading to hyponatraemia occur in 6-31% of patients taking carbamazepine. This is more common in the elderly. Sedation is frequent and can be significant in amount. Early signs of carbamazepine toxicity include dizziness, ataxia, sedation, and diplopia.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 270.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 250.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 302.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 356.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 132.

The correct answer is: Lithium → Polyuria, Venlafaxine → Sweating, Carbamazepine in the elderly → Hyponatraemia

Question **36**

Not answered

Marked out of 5.00

Flag question

Treatments in the elderly

Identify the treatment in each of the following descriptions used for various conditions in older patients.

This treatment, when combined with SSRIs, can induce serotonin syndrome — Choose...

This drug inhibits mucosal secretions and relaxes involuntary muscles — Choose...

Choose...
Lemon balm
St. John's wort
Gingko biloba
Aromatherapy
Atropine
Vitamin E
Lemon grass
Primrose oil

	Choose...
Used for the treatment of agitation in dementia	Choose...
Used for the treatment of cognitive symptoms of dementia	Choose...
Usually administered by skin massage to relieve tension, increase circulation and aid relaxation	Choose...

Explanation:

St. John's wort (SJW) is a potent inducer of intestinal and hepatic CYP3A4, CYP2C9, CYP2c39, CYP2E1, and intestinal p-glycoprotein. Hyperforin is responsible for this effect. The hyperforin content of SJW preparations varies 50-fold, which will result in a different propensity for drug interactions between brands. Serotonin syndrome has been reported when SJW was taken together with sertraline, paroxetine, nefazodone, and the triptans (a group of serotonin agonists used to treat migraine). SJW should not be taken with drugs that have a predominantly serotonergic action.

Atropine is a muscarine antagonist that blocks the actions of acetylcholine in the parasympathetic nervous system, paralysing smooth muscles. Atropine solution, given sublingually or used as a mouthwash, is sometimes used to manage hypersalivation and excess respiratory secretions in the terminally ill.

Recent NICE guidelines for dementia noted that there is evidence of some benefits across a range of domains from exercise, aromatherapy, music therapy, mindfulness, and animal assisted therapy. However, further trials are needed to gather convincing evidence that aromatherapy (or exposure to fragrant plant oils) is beneficial as a means of alleviating distressed behaviours in dementia. Peppermint oil is a relaxant of intestinal smooth muscle and does not appear to have an effect on cognition.

Although gingko biloba appears to be safe with no excess adverse effects compared with placebo, there is no convincing evidence that it is efficacious for dementia and cognitive impairment. Gingko's clinical benefit in dementia or cognitive impairment is somewhat inconsistent and unconvincing.

The essential oils used in aromatherapy are most commonly delivered through electric diffusers and vaporisers or massaged into the skin: thus the oil evaporates and the aroma stimulates the olfactory sense. The aromas used are generally experienced as pleasant and so the immediate effect may be a positive emotional response.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 418, 638.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

National Institute for Health and Clinical Excellence. The NICE guideline on the assessment, management and support for people living with dementia and their carers. 2018.

Ball EL et al. Aromatherapy for dementia. Cochrane Database of Systematic Reviews. 2020(8).

The correct answer is: This treatment, when combined with SSRIs, can induce serotonin syndrome → St. John's wort, This drug inhibits mucosal secretions and relaxes involuntary muscles → Atropine, Used for the treatment of agitation in dementia → Aromatherapy, Used for the treatment of cognitive symptoms of dementia → Gingko biloba, Usually administered by skin massage to relieve tension, increase circulation and aid relaxation → Aromatherapy

Question **37**
Not answered
Marked out of

Theories of drug action & side effects
Select one option for each description below.

3.00

Choose...
5HT2A stimulation
Alpha-1 blockade
Alpha 2 blockade
D2 receptor antagonism
Downregulation of alpha-2 adrenergic receptors
D1 receptor stimulation
CYP450 enzyme inhibition
CYP450 enzyme induction
Skeletal muscle calcium channel blockade
Direct ovarian stimulation via LH/FSH
5HT1A antagonism

A 34-year-old patient is recently started on risperidone. He develops dizziness on the mornings and has had two falls since the initiation of risperidone. He is not on any co-prescribed psychotropics. Choose...

A patient with bipolar disorder type 2 is treated with carbamazepine. She becomes pregnant in spite of adhering strictly to her oral contraceptive prescription. Choose...

A patient attending a Regional Affective Disorder Services has been prescribed lithium to augment the effect of fluoxetine for depression. She experiences tremors, hyperthermia, sweating and confusion. On examination, hyperreflexia is noted. Choose...

Explanation:

Risperidone's blockade of alpha-1 adrenergic receptors can cause dizziness, hypotension, and syncope.

Psychiatric drugs that are CYP3A4 inducers (e.g. carbamazepine, St John's Wort, phenytoin, topiramate) reduce the bioavailability and hence contraceptive efficacy of combined hormonal contraceptives, the progesterone-only pill, and the subdermal implant both during use and for up to 28 days after stopping enzyme-inducer treatment. Thus, patients on enzyme inducers should be encouraged to use intrauterine contraception or injectables.

Multiple pharmacologic agents are known to cause serotonin syndrome. Most frequently implicated are 3 classes of psychotropic drugs that boost serotonin in the synaptic cleft: SSRIs, tricyclic antidepressants, and MAOIs. Other common psychotropic drugs include certain amphetamines such as phentermine and select opiate receptor-binding agents such as oxycodone, meperidine, and tramadol. The antiemetic ondansetron, the antihistamine chlorpheniramine, and the mood stabiliser lithium have also been associated with increased serotonin levels.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 688

Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 314.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 855.

Patel DD, Galarneau D. Serotonin syndrome with Fluoxetine: two case reports. Ochsner Journal. 2016 Dec 21;16(4):554-7.

The correct answer is: A 34-year-old patient is recently started on risperidone. He develops dizziness on the mornings and has had two falls since the initiation of risperidone. He is not on any co-prescribed psychotropics. → Alpha-1 blockade, A patient with bipolar disorder type 2 is treated with carbamazepine. She becomes pregnant in spite of adhering strictly to her oral contraceptive prescription. → CYP450 enzyme induction, A patient attending a Regional Affective Disorder Services has been prescribed lithium to augment the effect of fluoxetine for depression. She experiences tremors, hyperthermia, sweating and confusion. On examination, hyperreflexia is noted. → 5HT2A stimulation

Question **38**

Not answered

Drug interactions with combination therapies

For each of the following drug combinations, choose the most characteristic adverse effect from the

Marked out of 4.00

Choose...
Gynaecomastia
Dry mouth
Tardive dyskinesia
Hypersalivation
Presbyopia
Absence seizure
Rhinorrhoea
Anaemia
Fever

list above.

A combination of imipramine and chlorpromazine is prescribed for depression with psychosis — Choose...

A combination of tranylcypromine and clomipramine prescribed for treatment-resistant depression at a specialized regional affective disorders unit — Choose...

A combination of risperidone and clozapine prescribed for treatment-resistant schizophrenia — Choose...

Explanation:

Tricyclic antidepressants, including imipramine, commonly cause anticholinergic adverse effects such as dry mouth. Chlorpromazine's blockade of muscarinic cholinergic receptors can also cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus.

Multiple pharmacologic agents are known to cause serotonin syndrome. Most frequently implicated are 3 classes of psychotropic drugs that boost serotonin in the synaptic cleft: SSRIs, tricyclic antidepressants (such as clomipramine), and MAOIs (such as tranylcypromine). Other common psychotropic drugs include certain amphetamines such as phentermine and select opiate receptor-binding agents such as oxycodone, meperidine, and tramadol. The antiemetic ondansetron, the antihistamine chlorpheniramine, and the mood stabiliser lithium have also been associated with increased serotonin levels.

Clozapine is well known to be causally associated with hypersalivation. Other antipsychotics, such as risperidone, have also been associated with hypersalivation, but to a lesser extent. Amisulpride has been found to be effective in treating clozapine-induced sialorrhea. Both amisulpride and quetiapine, when added to clozapine, have been shown to be effective in reducing clozapine-induced hypersalivation, and may allow dose reduction of clozapine.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 674.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 154.

Patel DD, Galarneau D. Serotonin syndrome with Fluoxetine: two case reports. Ochsner Journal. 2016 Dec 21;16(4):554-7.

Gupta S et al. Management of clozapine-induced sialorrhoea. BJPsych Advances. 2020 Mar;26(2):106-8.

The correct answer is: A combination of imipramine and chlorpromazine is prescribed for depression with psychosis → Dry mouth, A combination of tranylcypromine and clomipramine prescribed for treatment-resistant depression at a specialized regional affective disorders unit → Fever, A combination of risperidone and clozapine prescribed for treatment-resistant schizophrenia → Hypersalivation

Question **39**

Not answered

Marked out of 4.00

Flag question

Receptor mechanisms

For the following side effects exhibited by the drugs, identify the receptor action responsible for it

A 40-year-old woman on fluoxetine developed nausea and vomiting — Choose...

A 65-year-old woman complained of excessive sedation on taking trazodone — Choose...

A 33-year-old gentleman suffered from dry mouth on taking

Choose...
H1- antagonism
M1 receptor blockade
5-HT1 agonism
H2- antagonism
5-HT3 agonism
5-HT3 antagonism
5-HT1 antagonism
M2 receptor blockade
H3- antagonism
M3 receptor blockade
5-HT2 agonism

amitriptyline — Choose...

A 23-year-old woman has developed weight gain and is overly sedated on taking mirtazapine — Choose...

Explanation: SSRIs such as fluoxetine cause GI side-effects due to 5HT3 receptor stimulation. Trazodone causes sedation through H1 receptor antagonism. It can also induce priapism through alpha one receptor antagonism. Dry mouth is an anticholinergic side effect of tricyclics like amitriptyline and is due to M3 receptor blockade. Mirtazapine can cause sedation and weight gain through H1 receptor antagonism.

The correct answer is: A 40-year-old woman on fluoxetine developed nausea and vomiting → 5-HT3 agonism, A 65-year-old woman complained of excessive sedation on taking trazodone → H1-antagonism, A 33-year-old gentleman suffered from dry mouth on taking amitriptyline → M3 receptor blockade, A 23-year-old woman has developed weight gain and is overly sedated on taking mirtazapine → H1- antagonism

Question **40**

Not answered

Marked out of 4.00

Choose...
Escitalopram
Mirtazapine
Fluoxetine
Paroxetine
Trazadone
Minocycline
Citalopram
Venlafaxine
Agomelatine
Reboxetine

Antidepressants

For each of the following descriptions choose the most appropriate antidepressant from the given list

Which of the following drugs is associated with a specific risk of hypertension? — Choose...

Which of the following drugs is selective for noradrenergic reuptake inhibition? — Choose...

Which of the following drugs is called as a noradrenergic and specific serotonergic antagonist? — Choose...

Stimulation of certain serotonergic receptors leading to the adverse effects such as insomnia, agitation, sexual dysfunction and nausea is seen commonly during the use of antidepressants. Which of the following antidepressant has a low propensity for these effects? — Choose...

Explanation:

Depending on the dose, venlafaxine has different degrees of inhibition of serotonin reuptake (most potent and robust even at low doses) versus norepinephrine reuptake (moderate potency and robust only at higher doses). Adding norepinephrine transporter inhibition likely accounts for two side effects of venlafaxine in some patients: sweating and elevated blood pressure.

Reboxetine is a selective norepinephrine reuptake inhibitor.

Mirtazapine is a noradrenergic and specific serotonergic antidepressant. It is a multifunctional drug with five principal mechanisms of action: 5HT-2A, 5HT-2C, 5HT-3, alpha-2 adrenergic, and H1 histamine antagonism. It has a low propensity for nausea/vomiting and sexual dysfunction. A common side effect is sedation. 5HT-2A antagonism improves sleep, especially slow-wave sleep.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 302, 308-309.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 509-510, 681.

The correct answer is: Which of the following drugs is associated with a specific risk of hypertension? → Venlafaxine, Which of the following drugs is selective for noradrenergic reuptake inhibition? →

Reboxetine, Which of the following drugs is called as a noradrenergic and specific serotonergic antagonist? → Mirtazapine, Stimulation of certain serotonergic receptors leading to the adverse effects such as insomnia, agitation, sexual dysfunction and nausea is seen commonly during the use of antidepressants. Which of the following antidepressant has a low propensity for these effects? → Mirtazapine

Question **41**

Not answered

Marked out of 4.00

Partial agonism

Identify the predominant receptor level of action for the following partial agonistic drugs.

Buspirone Choose...

Clobazam Choose...

Buprenorphine Choose...

Aripiprazole Choose...

Choose...
D-5 receptor
5 HT1A receptor
5HT3 receptor
Mu opioid receptor
Omega opioid receptor
D-2 receptor
5HT2 receptor
Kappa opioid receptor
D-1 receptor
Benzodiazepine receptor

Explanation:

Buspirone is classified as an azapirone (or azaspirone) and is chemically distinct from other psychotropic agents. It is a 5-HT1A partial agonist.

The mechanism of action of clobazam is similar to other benzodiazepines. However, clobazam is believed to be a partial agonist rather than a non-selective full receptor agonist.

Buprenorphine is a partial agonist at mu-opioid receptors.

Aripiprazole is a D2 receptor partial agonist. It is also a partial agonist at 5-HT1A receptors, a high affinity antagonist at 5-HT2A receptors, and a low-/moderate affinity antagonist at H1 and alpha-1 receptors. It has no anticholinergic effect.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 672.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 176, 478.

Pernea M, Sutcliffe AG. Clobazam and its use in epilepsy. Pediatric Reports. 2016 Jun;8(2):34-8.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 212.

The correct answer is: Buspirone → 5 HT1A receptor, Clobazam → Benzodiazepine receptor, Buprenorphine → Mu opioid receptor, Aripiprazole → D-2 receptor

Question **42**

Not answered

Marked out of 6.00

Flag question

Antidementia drugs

For the antidementia drugs given below, identify their mode of action from list shown here:

Accessory effect of galantamine Choose...

Memantine Choose...

Accessory effect of rivastigmine Choose...

Choose...
Affects dopamine transmission
Nicotine receptor antagonist
Nicotine receptor agonist
Butyrylcholinesterase inhibitor
NMDA receptor agonist
NMDA receptor antagonist
Irreversible competitive inhibitor of AchEIs
Irreversible non-competitive inhibitor of AchEIs

Explanation: Binding to the Acetyl Cholinesterase enzyme (AChE) sites may be either reversible or irreversible, and may be competitive or noncompetitive with acetylcholine. Rivastigmine is a non-competitive inhibitor of both acetyl cholinesterase and butyrylcholinesterase. Galantamine is a competitive drug. It is a competitive and reversible inhibitor of acetyl cholinesterase. It is also a nicotine modulator. Memantine is an NMDA receptor antagonist and a weak 5-HT3 receptor antagonist.

Characteristic of cognitive enhancer	Donepezil	Rivastigmine	Galantamine	Memantine
Primary mechanism	AChE-I (selective and reversible)	AChE-I (reversible, non-competitive inhibitor)	AChE-I (competitive and reversible)	NMDA receptor antagonist (non-competitive)
Other mechanism	None	BuChE-I	Nicotine receptor agonist	5-HT3 receptor antagonist
Starting dose	5 mg daily	1.5 mg bd (oral) (or 4.6 mg/24 hours patch)	4 mg bd solution or 8 mg XL daily	5 mg daily
Usual treatment dose (and max dose)	10 mg daily	3-6 mg bd (oral) or 9.5 mg/24 hours patch	8-12 mg bd solution or 16-24 mg XL daily	20 mg daily or (10 mg bd)
Adverse effects	Diarrhoea, nausea, and headache most common	Anorexia, dizziness, nausea, vomiting, and diarrhoea most common	Nausea and vomiting most common	Dizziness, headache, fatigue, diarrhoea, and gastric pain most common
Half-life (hours)	~70 (permitting once daily dosing)	~1 (oral) – active for 10 3.4 (patch)	7-8 (oral solution) 8-10 (XL capsules)	60-100
Metabolism	CYP3A4 CYP2D6 (minor)	Minimal involvement of CYP isoenzymes	CYP3A4 CYP2D6	Primarily non-hepatic
Drug-drug interactions	Yes	Interactions unlikely	Yes	Yes
Effect of food on absorption	None	Delays rate and extent of absorption	Delays rate but not extent of absorption	None
Other	Absorbed completely from the GI tract Fairly well-tolerated and widely used	More likely to cause GI adverse effects than donepezil due to more peripheral activity	More likely to cause GI and neuropsychiatric adverse effects than donepezil	Generally better tolerated than cholinesterase inhibitors

AChE-I = acetylcholinesterase inhibitor
BuChE-I = butyrylcholinesterase inhibitor

Adapted from Maudsley Prescribing Guidelines, 2021. pp. 605-6.

SPMM COURSE

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 605-606.

The correct answer is: Accessory effect of galantamine → Nicotine receptor agonist, Memantine → NMDA receptor antagonist, Accessory effect of rivastigmine → Butyrylcholinesterase inhibitor

Question **43**

Not answered

Marked out of 6.00

Choose...
Blurred vision
Hypothyroidism
Diastolic hypertension
QTc prolongation
Word finding difficulties
Akathisia
Hypersalivation
Bradycardia
Polydipsia
Nasal stuffiness

Adverse effects

For each of the following drugs, identify the most characteristic side effect associated with its therapeutic use

Galantamine — Choose...

Venlafaxine — Choose...

Topiramate — Choose...

Lithium — Choose...

Aripiprazole — Choose...

Explanation:

Excess cholinergic stimulation with acetylcholinesterase inhibitors (such as galantamine) can lead to nausea, vomiting, dizziness, insomnia, and diarrhoea. Acetylcholinesterase inhibitors can also be expected to have vagotonic effects on heart rate (i.e. bradycardia). Nausea and vomiting are the most common side effects of galantamine, but bradycardia can occur and may have potential to be more serious.

Depending on the dose, venlafaxine has different degrees of inhibition of serotonin reuptake (most potent and robust even at low doses) versus norepinephrine reuptake (moderate potency and robust only at higher doses). Adding norepinephrine transporter inhibition likely accounts for two side effects of venlafaxine in some patients: sweating and elevated blood pressure.

Topiramate has been widely reported to cause adverse cognitive effects specifically related to working memory and verbal fluency in healthy volunteers and patients with epilepsy and migraine.

Common side effects of lithium include mild gastrointestinal upset, fine tremor, polyuria, and polydipsia. In the longer term, lithium increases the risk of hypothyroidism.

Aripiprazole lacks the pharmacological properties normally associated with sedation, namely muscarinic cholinergic and H1 histamine antagonist properties, and thus is generally not sedating. Aripiprazole is generally well-tolerated with little weight gain, but some patients experience akathisia.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 250, 610.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 239, 302, 327.

Ahmed GF et al. Pharmacokinetic–pharmacodynamic modelling of intravenous and oral topiramate and its effect on phonemic fluency in adult healthy volunteers. British journal of clinical pharmacology. 2015 May;79(5):820-30.

The correct answer is: Galantamine → Bradycardia, Venlafaxine → Diastolic hypertension, Topiramate → Word finding difficulties, Lithium → Hypothyroidism, Aripiprazole → Akathisia

Question **44**

Not answered

Marked out of 5.00

Teratogenicity of psychotropics

Choose the most likely offending agent for each of the following teratogenic effects from the given list.

Newborn with cleft palate and features similar to foetal alcohol syndrome — Choose...

Newborn with symptoms of antidepressant discontinuation reaction — Choose...

Children with d evelopmental delay, low IQ, and features suggestive of autism spectrum — Choose...

Choose...
Carbamazepine
Zopiclone
Paroxetine
Reboxetine
Lithium
Mirtazapine
Diazepam
Valproate

Explanation:

While the teratogenic effects of benzodiazepines appear to be lower in large cohorts than what has been reported in early case-control studies (including a two-fold increase in cleft palate risk), the risk of overall and heart defects is higher in mothers who receive higher doses. A foetal alcohol syndrome like presentation has also been reported.

When antidepressants are taken during pregnancy, the neonate may experience discontinuation symptoms, which are usually mild, such as agitation and irritability, or rarely respiratory distress and convulsions (with SSRIs). The risk is assumed to be particularly high with short half-life drugs such as paroxetine and venlafaxine. Continuing to breastfeed and then 'weaning' by switching to mixed (breast/bottle) feeding may help reduce the severity of reactions.

There is strong evidence for increased risk of congenital and major congenital malformations with valproate (as many as 11% of exposed children), including cleft lip and/or palate, neural tube defects (risk of spina bifida 1-2%), congenital heart anomalies, radial ray defects, ophthalmological and genitourinary anomalies, developmental delay (delayed walking and talking, memory problems,

difficulty with speech and language), low IQ, and autism.

Ref: Noh Y, Lee H, Choi A, Kwon JS, Choe SA, Chae J, Kim DS, Shin JY. First-trimester exposure to benzodiazepines and risk of congenital malformations in offspring: A population-based cohort study in South Korea. PLoS medicine. 2022 Mar 2;19(3):e1003945.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 689.

McAllister-Williams RH et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. Journal of Psychopharmacology. 2017 May;31(5):519-52.

The correct answer is: Newborn with cleft palate and features similar to foetal alcohol syndrome → Diazepam, Newborn with symptoms of antidepressant discontinuation reaction → Paroxetine, Children with developmental delay, low IQ, and features suggestive of autism spectrum → Valproate

Question **45**

Not answered

Marked out of 2.00

Adverse effects

Identify the type of side effects caused by antipsychotics, using descriptions given below

A 25-year-old gentleman with an acute episode of psychosis was started on trifluoperazine 3 days ago. He has now developed a high fever, rigidity, profuse sweating and confusion.

Choose...

A 69-year-old lady on long-term treatment with depixol injection presents with lip smacking, chewing and protrusion of the tongue.

Choose...

Choose...
Cerebellar ataxia
Akathisia
SIADH
Parkinsonism
Neuroleptic malignant syndrome
Tardive dyskinesia
Hyponatraemia
Serotonin syndrome
Acute dystonia
Cholinergic rebound syndrome

Explanation:

Neuroleptic malignant syndrome (NMS) is rare, occurring in approximately 1% of those treated with first-generation antipsychotics, but also in those treated with second-generation drugs. In addition, there have been reports of NMS following treatment with other medications, including valproate, lithium, and antidepressants (such as amoxapine, clomipramine, desipramine, phenelzine, trimipramine, and venlafaxine). Signs/symptoms include hyperthermia, muscular rigidity, confusion/agitation/altered level of consciousness, tachycardia, tachypnoea, hyper-/hypotension, diaphoresis/sialorrhea, tremor, incontinence/retention/obstruction, creatinine kinase/urinary myoglobin, leucocytosis, and metabolic acidosis.

Tardive dyskinesia is characterised by late onset (mean 7 years) involuntary, repetitive, purposeless movements occurring with long-term antipsychotic treatment. Peri-oral movements are the most common. Risk factors for tardive dyskinesia include chronic use of antipsychotics, change/cessation of chronic treatment, concomitant anticholinergic treatment, elderly (>60 years), female, organic disorder (e.g. dementia, intellectual disability, epilepsy), previous head injury, alcoholism, comorbid mood disorder, negative symptoms of schizophrenia, diabetes mellitus, history of previous drug-induced akathisia/Parkinsonism/dystonias, and concomitant use of predisposing drugs.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 1014, 1018.

The correct answer is: A 25-year-old gentleman with an acute episode of psychosis was started on trifluoperazine 3 days ago. He has now developed a high fever, rigidity, profuse sweating and confusion. → Neuroleptic malignant syndrome, A 69-year-old lady on long-term treatment with depixol injection presents with lip smacking, chewing and protrusion of the tongue. → Tardive dyskinesia

Question **46**

Not answered

Marked out of 5.00

Receptor mechanisms

Identify the drugs & receptors that display the following features

Partial agonist — Choose...

Competitive antagonist — Choose...

Noncompetitive antagonist — Choose...

Choose...
Propranalol at beta adrenergic receptors
Phencyclidine at NMDA receptors
Buprenorphine
MAOIs
Ketamine at NMDA receptors
Benzodiazepines at GABA A complex

Explanation:

Buprenorphine is a partial agonist at mu-opioid receptors.

Propranolol is a nonselective, competitive antagonist at beta-adrenergic receptors. Propranolol exerts its effects primarily by blocking the action of the endogenous catecholamines, epinephrine and norepinephrine, at beta adrenergic receptors. The magnitude of the effect of propranolol is dependent on the degree of sympathetic tone. Propranolol also blocks the effects of drugs with beta adrenergic receptor agonist activity.

Ketamine is a non-competitive open channel inhibitor of the NMDA receptor. Specifically, it binds to the phencyclidine site of the NMDA receptor.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 478.

Bylund DB. Propranolol. In: Reference Module in Biomedical Sciences. Elsevier, 2015.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 383.

The correct answer is: Partial agonist → Buprenorphine, Competitive antagonist → Propranalol at beta adrenergic receptors, Noncompetitive antagonist → Ketamine at NMDA receptors

Question **47**

Not answered

Marked out of 4.00

Flag question

Chemical class of psychotropic drugs

Match each chemical class below with the most appropriate drug from the given list:

Aliphatic phenothiazine — Choose...

Butyrophenones — Choose...

Piperazine derivatives — Choose...

Piperidine derivatives — Choose...

Choose...
Quetiapine
Chlorpromazine
Ziprasidone
Molindone
Thioridazine
Loxapine
Risperidone
Haloperidol
Zuclopenthixol
Trifluoperazine

Explanation:

Aliphatic phenothiazines are chlorpromazine-like drugs with mainly anti-adrenergic and antihistaminergic side effects, including pronounced sedation, moderate antimuscarinic effects, and moderate EPSEs. This group includes chlorpromazine, levomepromazine, and promazine.

Butyrophenones are similar to piperazine phenothiazines in that they are high potency and tend to produce troublesome EPSEs. This group includes haloperidol and benperidol.

Piperazine phenothiazines are trifluoperazine-like drugs with mainly anti-dopaminergic side effects.

These drugs are potent antipsychotics but tend to produce troublesome EPSEs, particularly at higher doses, with limited sedative properties. This group includes trifluoperazine, fluphenazine, and perphenazine.

Piperidine phenothiazines are thioridazine-like drugs with mainly antimuscarinic side effects and fewer EPSEs than the previous groups. This group includes thioridazine and pericyazine.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 208-209.

The correct answer is: Aliphatic phenothiazine → Chlorpromazine, Butyrophenones → Haloperidol, Piperazine derivatives → Trifluoperazine, Piperidine derivatives → Thioridazine

Question **48**

Not answered

Marked out of 5.00

Chemical class of psychotropic drugs

Match each drug below with the most appropriate chemical class from the given list:

Zopiclone Choose...

Imipramine Choose...

Nortryptyline Choose...

Phenelzine Choose...

Zolpidem Choose...

Choose...
Butyrophenone
Triazolopyridine
Imidazopyridine
Hydrazine derivative
Piperidine derivative
Tertiary amine
Aliphatic phenothiazine
Piperazine derivative
Cyclopyrrolone
Secondary amine

Explanation:

Zopiclone is a cyclopyrrolone and zolpidem is an imidazopyridine.

Tertiary amines include amitriptyline and imipramine. Secondary amines include amoxapine, desipramine, and nortriptyline. Nortriptyline and desipramine (not available in the UK) are less anticholinergic and hypotensive than amitriptyline and imipramine (respectively, their tertiary amine parent molecules).

Phenelzine is an MAOI and a hydrazine derivative. Isocarboxazid is also a hydrazine derivative. Both are less stimulating than other MAOIs.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 406, 686.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 283.

The correct answer is: Zopiclone → Cyclopyrrolone, Imipramine → Tertiary amine, Nortryptyline → Secondary amine, Phenelzine → Hydrazine derivative, Zolpidem → Imidazopyridine

Question **49**

Mechanism of action (1)

Identify the drug that matches each of the following mechanisms of action.

Alpha autoreceptor and selective serotonergic antagonist. Choose...

Serotonin and norepinephrine reuptake inhibitor. Choose...

An inhibitory metabolite of glutamate Choose...

Choose...
Duloxetine
Lactate
Lithium
Sildenafil
Fluoxetine
Aspartate
Sodium mono glutamate
GABA
Mirtazapine
Glutamate

	Choose...
Phosphodiesterase-5 inhibitor	Choose...

Explanation:

Mirtazapine is a noradrenergic and specific serotonergic antidepressant. It is a multifunctional drug with five principal mechanisms of action: 5HT-2A, 5HT-2C, 5HT-3, alpha-2 adrenergic, and H1 histamine antagonism.

Serotonin-norepinephrine reuptake inhibitors (SNRIs) combine robust serotonin transporter (SERT) inhibition with varying degrees of inhibition of the norepinephrine transporter (NET). Duloxetine is an SNRI that is characterised pharmacologically by slightly more potent SERT than NET inhibition.

GABA is the principal inhibitory neurotransmitter in the brain, and normally serves an important regulatory role in reducing the activity of many neurons. GABA is produced, or synthesised, from the amino acid glutamate (glutamic acid) via the actions of the enzyme glutamic acid (glutamate) decarboxylase.

Sildenafil is a phosphodiesterase-5 inhibitor.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 255, 298-302, 308-309.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 729.

The correct answer is: Alpha autoreceptor and selective serotonergic antagonist. → Mirtazapine, Serotonin and norepinephrine reuptake inhibitor. → Duloxetine, An inhibitory metabolite of glutamate → GABA, Phosphodiesterase-5 inhibitor → Sildenafil

Question **50**

Not answered

Marked out of

History of pharmacology (1)

For each of the following historical figures choose one most appropriate discovery made from the above list

Choose...
Lithium for mania
Sildenafil for erectile dysfunction
Imipramine for depression
Chlorpromazine for psychosis
Carbamazepine for mood disorders
Electroconvulsive therapy
Haloperidol for schizophrenia
Clozapine for resistant schizophrenia

Osterloh	Choose...
Takezaki & Hanaoka	Choose...
Cade	Choose...
Kuhn	Choose...
Cereletti & Bini	Choose...

Explanation:

Sildenafil was synthesised by a group of pharmaceutical chemists working at Pfizer's Kent research facility in England. It was initially studied for use in hypertension and angina pectoris. Phase I clinical trials under the direction of Ian **Osterloh** suggested that the drug had little effect on angina, but that it could induce marked penile erections. Pfizer therefore decided to market it for erectile dysfunction, rather than for angina.

Carbamazepine is a widely used anticonvulsant that was first evaluated as a potential treatment in manic depressive psychosis by **Takezaki and Hanaoka**, and Okuma et al. in the 1970s. There have

since been many studies demonstrating the efficacy of carbamazepine in treating the acuate manic and depressive symptoms of bipolar disorder, as well as in prophylaxis.

The use of lithium in psychiatry goes back to the mid-19 th century. Early work, however, was soon forgotten, and John **Cade** is credited with reintroducing lithium to psychiatry for mania in 1949.

In 1955, Ronald **Kuhn**, a Swiss psychiatrist, examined the antidepressant effects of an unknown Geigy compound designated G22355 (later named imipramine). The antipsychotic effects of chlorpromazine had been recently discovered and drug companies, including Geigy, were collaborating with hospital-based psychiatrists like Kuhn to come up with other compounds that would have the antipsychotic effects of chlorpromazine. Kuhn noted that although one of these compounds, G22355, did not reliably improve psychotic symptoms, some of the schizophrenia patients became hypomanic from it and G22355 seemed to alleviate depression in a few schizophrenia patients who had prominent depressive symptoms. With the encouragement and support of Geigy, Kuhn undertook a study of the antidepressant effects of G22355.

Two Italian psychiatrists, **Ugo Cerletti and Lucio Bini** were interested in the use of electricity to induce seizures. In 1938, they conducted the first electrical induction of a series of seizures in a catatonic patient and produced a successful treatment response.

Ref: Hirschfeld RM, Kasper S. A review of the evidence for carbamazepine and oxcarbazepine in the treatment of bipolar disorder. International Journal of Neuropsychopharmacology. 2004 Dec 1;7(4):507-22.

Shorter E. The history of lithium therapy. Bipolar disorders. 2009 Jun;11:4-9.

Brown WA, Rosdolsky M. The clinical discovery of imipramine. American Journal of Psychiatry. 2015 May 1;172(5):426-9.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 729-730.

The correct answer is: Osterloh → Sildenafil for erectile dysfunction, Takezaki & Hanaoka → Carbamazepine for mood disorders, Cade → Lithium for mania, Kuhn → Imipramine for depression, Cereletti & Bini → Electroconvulsive therapy

Finish review

PREVIOUS ACTIVITY
Adverse effects

NEXT ACTIVITY
EMIs Pharmacology Part 2

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